Engine Model(s): PT6A

72-00-00

ENGINE - INSPECTION

1. General

A. The inspection criteria for the various sections of the engine are summarized in Table 601. Detailed inspection procedures are provided, where applicable, in the relevant INSPECTION/CHECK sections of subject chapters in this manual.

B. The inspection procedures detailed are considered a normal function of operating organizations and are intended as a guide for minimum inspection and maintenance requirements.

C. ROUTINE inspection coincides with the daily or preflight airframe inspection. MINOR inspection is coincidental with a typical airframe zone inspection. The intervals at which these inspections are performed may be altered by the aircraft manufacturers maintenance program and approved by the operators local airworthiness authority.

D. Engines maintained in compliance with the aircraft manufacturer's approved maintenance program must be inspected as necessary by Condition Monitoring in accordance with the Operator/Local Authority agreed plan. Engines operating in salt-laden or harsh atmospheres must be subjected to additional inspections for corrosion and compressor erosion.

E. Hot section inspection frequencies are outlined in the applicable service bulletin (Ref. DESCRIPTION AND OPERATION, Approved Service Bulletins) and will be updated periodically as dictated by service experience and time-between-overhauls (TBO). If, in the light of experience, it becomes necessary to modify procedures, these will be included in subsequent revisions to this manual.

F. Retaining rings (spirolox, etc.) used throughout the engine are to be inspected for distortion and general condition. Distorted rings are not acceptable and must be replaced (Ref. 70-00-00, MAINTENANCE PRACTICES).

2. Consumable Materials

The consumable materials listed below are used in the following procedures.

Item No.	Name
PWC04-001	Grease
PWC11-027	Solvent, Petroleum
PWC11-031	Cleaner, Engine
PWC15-011	Inhibitor, Rust
PWC15-015	Compound, Corrosion Preventative
PWC15-016	Compound, Corrosion Inhibitor

3. Special Tools

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The special tools listed below are used in the following procedures.

Tool No.	Name
PWC34910-101	Borescope Assembly
PWC34910-200	Guide Tube
PWC34913	Holding Fixture
PWC34941	Wrench

4. Fixtures, Equipment and Supplier Tools

Not Applicable

5. Inspection for Specific Operating Environments

A. General

Some inspection task intervals in Table 601, Periodic Inspection or other maintenance tasks (like engine wash) may be adjusted to account for a specific operating environment and are developed to improve engine reliability or to reduce operating cost.

For the purpose of adjusting the inspection intervals to the operating environment, three distinctive geographical regions are defined: sand & dust, polluted/sulfur-rich and salt-laden. Engines may be subjected to more than one of these environmental conditions simultaneously. The maintenance inspections contained in this section provide guidance to define the recommended tasks and intervals.

The sand/dust environment may cause heavy erosion to both the compressor and hot section components or obstruct the secondary air system, such as air-cooling passages of hot section components including vanes. Sand or dust can also accumulate within engine controls and accessories, which may lead to engine performance degradation or affect engine operation.

Operating in polluted or sulfur-rich environments can cause corrosion, oxidation, hot corrosion/erosion to the engine. Air pollution in adverse environments combined with chemical reactions of fuel and heat in the engine can result in severe damage to the engine's hot section, commonly referred to as sulfidation. Additionally, like the sand/dust environment, pollution contaminants can also lead to blockage of cooling holes or accumulate within engine controls and accessories, which may lead to engine performance degradation or affect engine operation.

Sea salt is one of the most globally abundant aerosols to which engines may be exposed during operation. The presence of salt haze, created by finely divided particles of sea salt in the air, usually derived from the evaporation of sea spray will find its way within the engine gas path, secondary air system and on external engine surfaces. The salt-laden environment will cause engine deterioration by corrosion and in the absence of preventive maintenance measures can lead to hot corrosion or sulfidation. Engine operation in salt-laden environment may also affect engine controls and accessories, which may lead to engine performance degradation or affect engine operation.

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Unprepared landing surfaces are typically surfaces comprised of gravel, turf, clay or hard packed soil mixtures. Seasonal temperature variation may influence the surface roughness and cause loss of compaction. Although not considered as a specific environment as described in this section, operators taking off and landing on unprepared surfaces should consider adapting their maintenance practices to those for a seasonal dust environment.

Additionally, operators should review and adjust (if required) their environmental maintenance tasks and intervals when an aircraft or engine changes its flight operations by considering prior and new operations.

P&WC may assist operators if required to review the specific operating environments and to provide maintenance recommendation.

B. Definitions

(1) Frequency of Operation in a Specific Environment

(a) The frequency of flight operation is defined as the total number of flights performed in a specific environment for a given period over the total number of flights of that same period within all environment types. For engines operating in a mix of environments, each environment frequency should be established.

(b) For the purpose of establishing the proper maintenance interval, the frequencies of flight operations in a specific environment are defined as follows:

- 80 to 100% - Continually
- 50 to 79% - Frequently
- 20 to 49% - Occasionally
- 0 to 19% - Rarely

(2) Specific Environments

(a) Sand/Dust Environments: For the purpose of determining targeted maintenance actions, the sand and dust environment is sub-classified into two zones (Ref. Fig. 601). Zone 1 represents a permanent arid environment and Zone 2 is an environment subject to dry seasons.

(b) Polluted/Sulfur-rich Environment: For the purpose of determining targeted maintenance actions in highly industrialized and populated regions, the annual average Air Quality Index (AQI) per the U.S Environmental Protection Agency (EPA) 2016 standard or equivalent is used as a reference. The environments presented over a 10 years average in Figure 602 are sub-classified into three zones: Zone 1: Moderate pollution (AQI>51); Zone 2: Severe pollution (AQI>101); Zone 3: Extreme pollution (AQI>151).

(c) Salt-laden Environment: Regions in the proximity of large bodies of salt water. For the purpose of determining targeted maintenance actions, these environments are sub-classified into three distinctive types of operations:

1 Coastal Operation - Operating within 27 Nautical Miles (50 Kilometers) of a coastline.

2 Island and Sea Surveillance/Maritime Low-level Flights - Operating above the sea at

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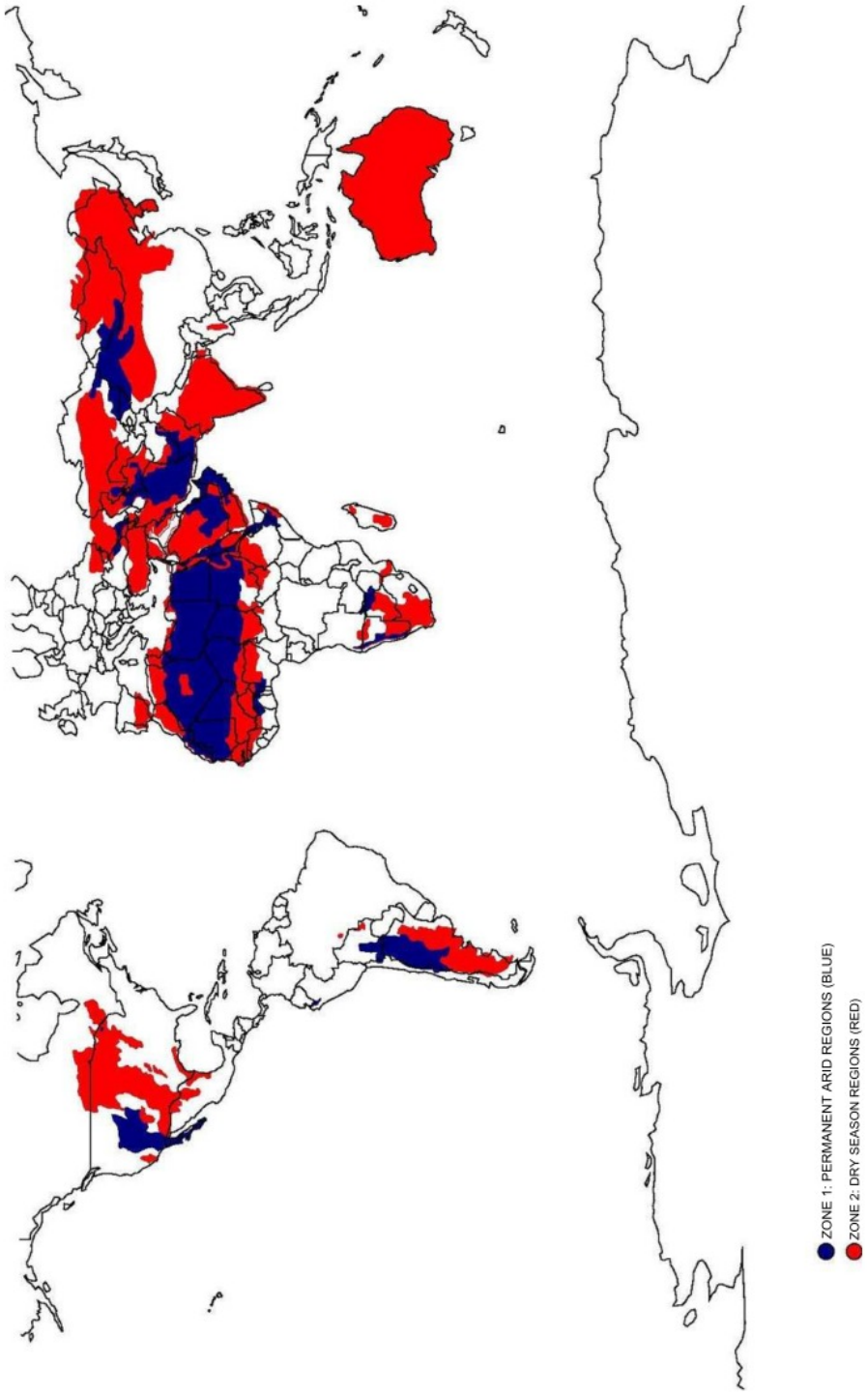
cruise altitudes below 10000 ft (3 km).

3 Sea Operation - Operating above the sea at cruise altitude greater than 10000 ft (3 km).

NOTE: Salt deposits or corrosion on various aircraft parts is further evidence that the engine is being operated in a salt-laden environment.

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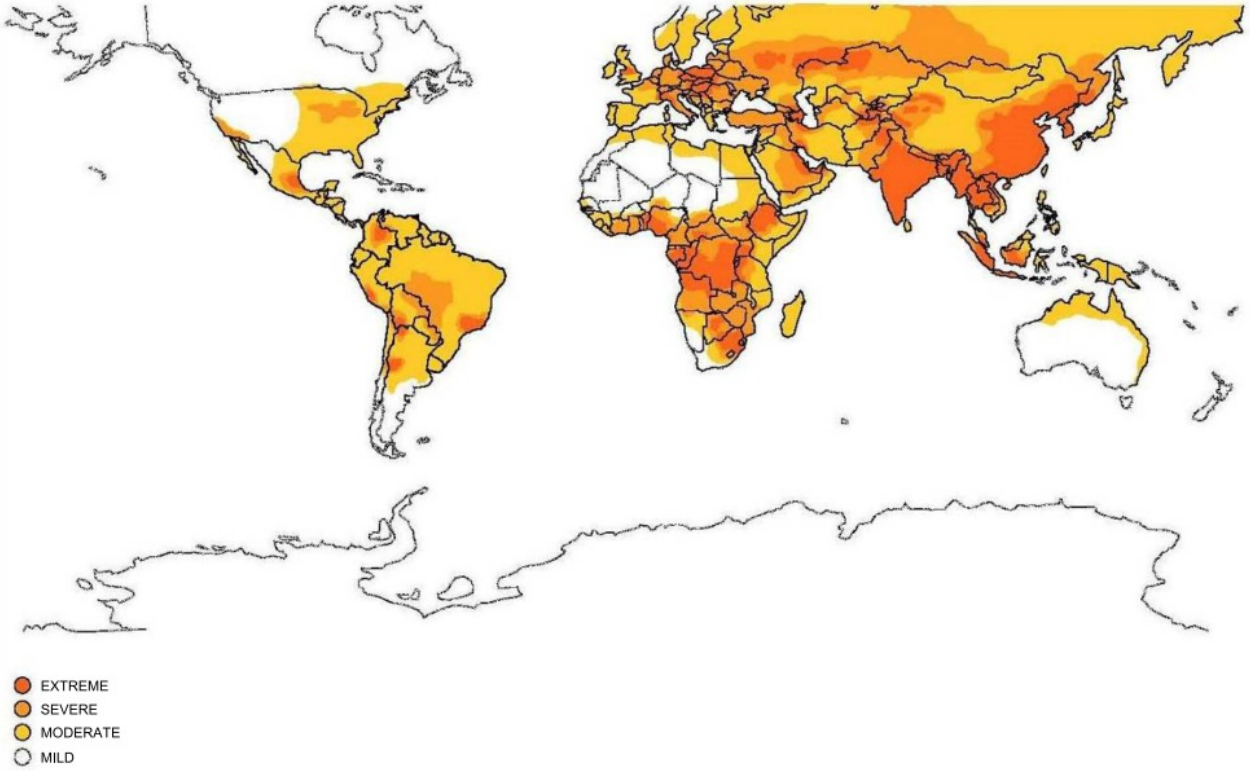
Figure 601 Sand/Dust Environment Map



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Figure 602 Pollution/Sulfur-rich Environment Map



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6. Corrosion

A. High Temperature Alloy Corrosion

High temperature alloy corrosion may be identified as metal loss or pitting, but more usually appears as local swelling or buildup due to greater volume occupied by nickel oxides. These corrosion products vary in color from black to green and in the advanced state there will be cracking and flaking, referred to as exfoliation.

Care should be taken to distinguish between corrosion buildup and possible light brown or rust colored deposits which are essentially harmless combustion by-products. The latter are usually more widespread over hot section components and while possibly affecting performance will not directly affect compressor turbine mechanical integrity.

B. Sulphidation

Sulphidation is a common name for a type of hot corrosion which can affect turbine area components of gas turbine engines.

Sulphates form at engine operating temperatures with sodium and sulphur present. Most aviation turbine fuels contain sulphur in sufficient amounts for sulphidation. Common sources of sodium are seawater, atmospheric pollutants and volcanic discharges.

Sulphidation attack most often affects the compressor turbine blades, but sulphidation of power turbine blades and of non-rotating parts such as shroud segments is not uncommon. Turbine blades are coated with a protective layer. Sulphidation will first degrade this coating and eventually lead to some loss of base alloy material. Post coating strip inspection of attacked blades may reveal that they are not suitable for repair by recoating.

In-service sulphidation may be characterized through the following four stages:

Stage 1 - Light Sulphidation (Initial Coating Deterioration): Slight roughening of the surface and localized breakdown of the protective surface layer. Substrate is not yet affected.

Stage 2 - Failure of the protective layer (Initial substrate corrosion): Roughness of surface is more marked as protective layer breakdown exposes the substrate to attack. Mechanical integrity is not yet affected.

Stage 3 - Severe sulphidation: Base material is attacked to significant depth, as evidenced by obvious buildup scale and some blistering. Progression to Stage 4 would accelerate. Integrity is now affected and parts are no longer serviceable.

Stage 4 - Catastrophic attack: Deep penetration of attack with large blisters of scale. Loss of structural material will lead to eventual component failure.

All stages of attack may be encountered, although stages 3 and 4 will normally be limited to those engines which are directly exposed to salt water.

Desalination wash using plain water with an antifreeze agent, if required, minimizes attack. During stages 1 and 2 the wash will dissolve and carry away sulfates on component surface.

Water used for washing must be free from contamination, in most cases drinking water is

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acceptable. Otherwise demineralized water should be used.

Wash frequency depends on the amount of contamination. Washing weekly is probably the minimum acceptable. Maritime areas may require washing daily or before each flight.

Recommended wash frequency is based on operators estimate of corrosion environment or refer to Chapter 71-00-00, CLEANING.

An alternative method of establishing wash frequency is to monitor blade condition using a borescope assembly (PWC34910-101). Wash schedule should be adjusted according to results.

If sulphidated blades continue in service, a borescope inspection program should be started. This program should continue until a reliable progression rate for the particular circumstances can be established and a blade life limit determined. The recommended inspection interval is 200 hours. Blade lives established this way should be reviewed periodically to account for possible changes in the factors affecting sulphidation.

Two points are important to consider when evaluating sulphidated blades:

Stage of corrosion attack.

Estimate of progression rate to stage 3.

At stage 3 mechanical integrity is jeopardized and blades should be removed from service. When estimating progression beyond stage 1 or 2 exposure to contamination and improvements in wash programs should be taken into account.

7. Corrosion Inhibition

A. Corrosion Inhibition Procedure

NOTE: 1. For engines operating in corrosive environments, it may be necessary to do more frequent periodic preventative maintenance including the application of corrosion inhibitors to the Accessory Gearbox Housing, Reduction Gearbox Housing (Front and Rear) and the Compressor Inlet Case.

NOTE: 2. As the corrosion inhibitor starts to evaporate or is washed away in service, reapply as required to determine the optimum period between applications.

NOTE: 3. Discontinue the use of (PWC09-003) if using a corrosion inhibitor.

NOTE: 4. Continue to follow the recommended external and internal engine wash schedule (Ref. Chap. 71-00-00) after use of corrosion inhibitors.

(1) Do a compressor desalination wash followed by a compressor turbine desalination wash (Ref. 71-00-00, Cleaning).

(2) Do an external engine wash (Ref. 71-00-00, Cleaning).

(3) For components listed below, remove corrosion in accordance with applicable Approved Repairs procedure:

- AGB Housing (Ref. 72-60-00, Approved Repairs)
- RGB Front and Rear Housings ((Ref. 72-10-00, Approved Repairs)

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- Compressor Inlet Case (Ref. 72-20-00, Approved Repairs)

WARNING: LET THE TEMPERATURE OF THE ENGINE DECREASE TO AMBIENT TEMPERATURE FULLY BEFORE YOU APPLY CORROSION INHIBITOR. IF YOU SPRAY CORROSION INHIBITOR ONTO A HOT SURFACE, THE CARRIER SOLVENT WILL QUICKLY EVAPORATE AND AN IGNITION CAN OCCUR IF AN IGNITION SOURCE IS THERE.

WARNING: DO NOT APPLY CORROSION INHIBITOR ON CONNECTORS, WIRING HARNESS, PROBES, RUBBER MATERIAL, ELECTRICAL EQUIPMENT OR ON HOT SECTION OF THE ENGINE.

WARNING: REFER TO THE MANUFACTURER'S MATERIAL SAFETY DATA SHEETS FOR INFORMATION SUCH AS HAZARDOUS INGREDIENTS, PHYSICAL/CHEMICAL PROPERTIES, FIRE, EXPLOSION, REACTIVITY, HEALTH HAZARD DATA, PRECAUTIONS FOR SAFE HANDLING, USE AND CONTROL MEASURES.

(4) Apply rust inhibitor (PWC15-011) or corrosion preventative compound (PWC15-015) or corrosion inhibitor compound (PWC15-016) or equivalent onto the painted surfaces with a spray or brush per the manufacturer's instructions.

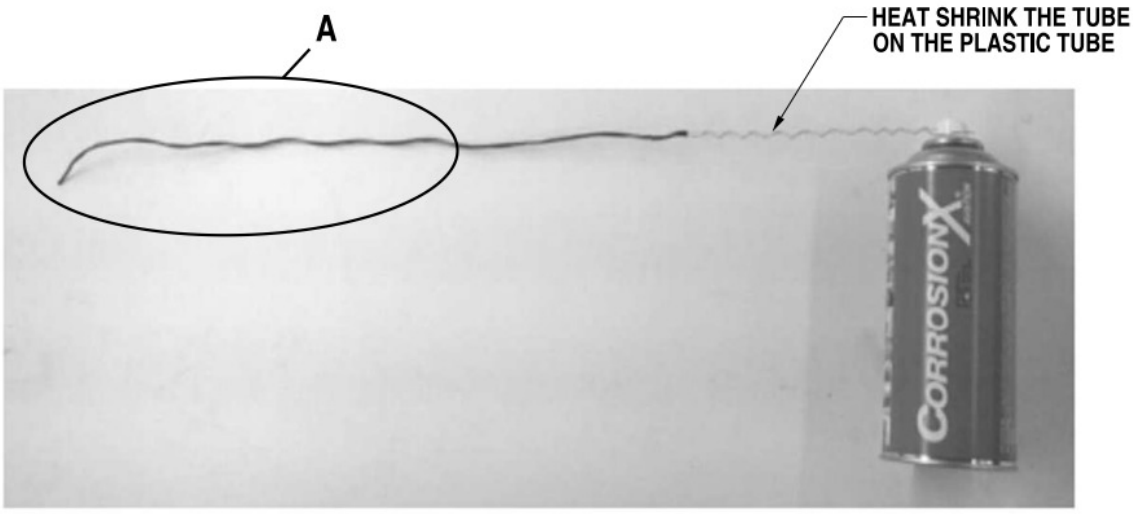
(5) Check to make sure that the surface is completely coated.

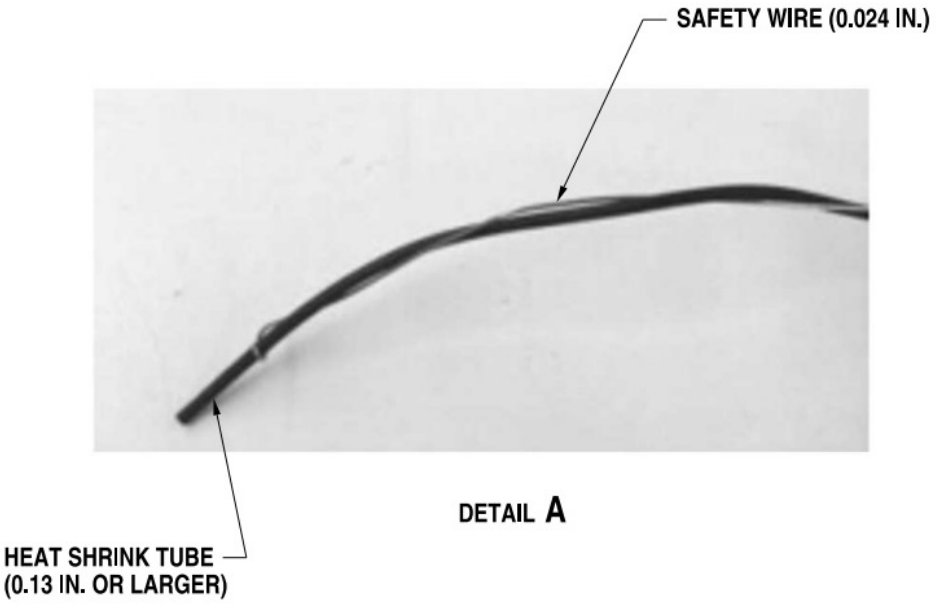
B. Air Inlet Case Corrosion Prevention (Ref. Fig. 603)

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Figure 603 Air Inlet Case Corrosion Prevention

(SHEET 1 OF 2)

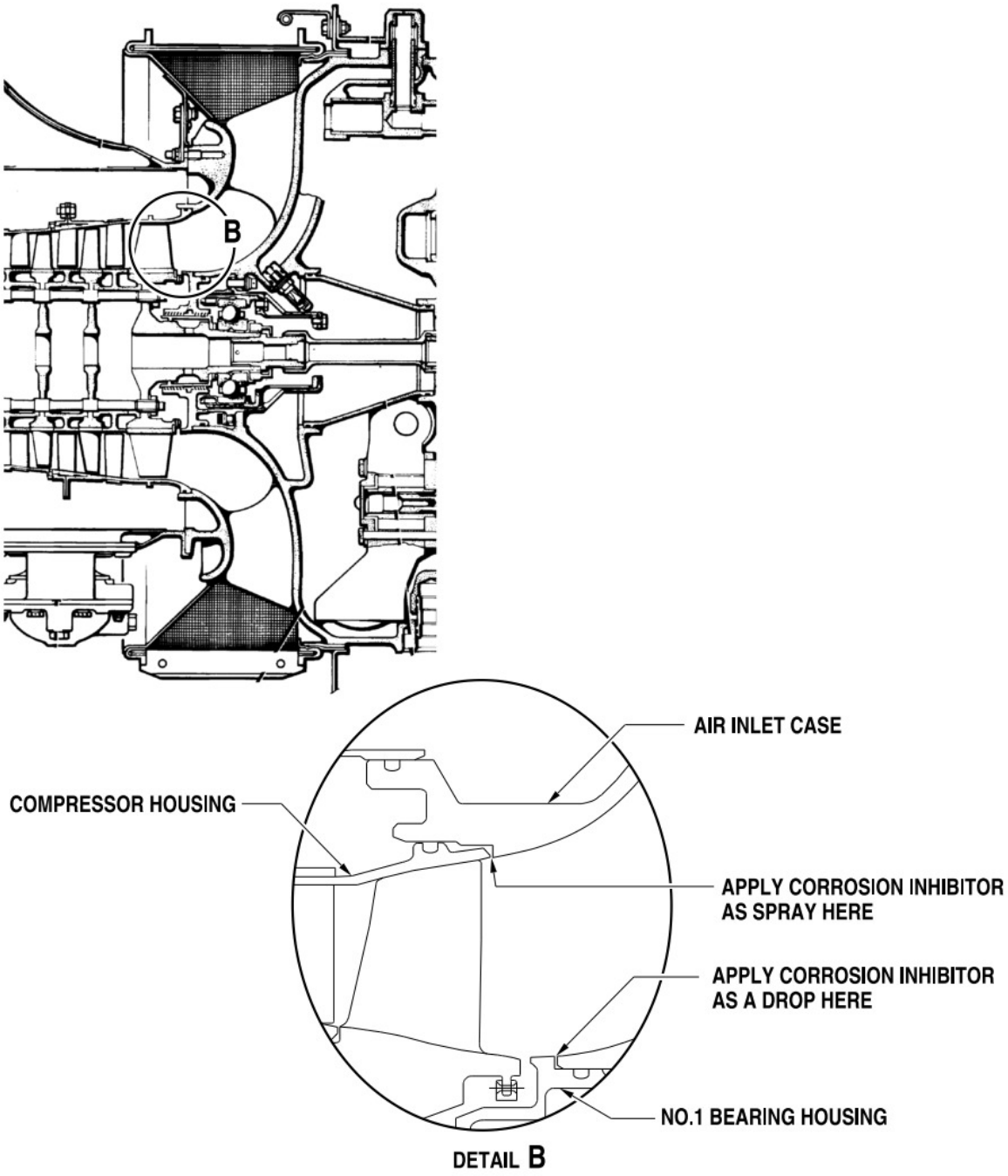




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(SHEET 2 OF 2)

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(1) Apply rust inhibitor (PWC15-011) or corrosion preventative compound (PWC15-015) or corrosion inhibitor compound (PWC15-016) or equivalent after each desalination wash.

NOTE: 1. Use of corrosion inhibitor will prevent the increase of corrosion without high maintenance work.

NOTE: 2. Use a larger diameter heat shrink tube when you apply corrosion inhibitor as a drop. Use a smaller diameter heat shrink tube to apply a spray of corrosion inhibitor solution.

NOTE: 3. Identify the location and area of corrosion on the inlet case, then apply the corrosion inhibitor as a drop or spray.

CAUTION: USE OF TOO MUCH CORROSION INHIBITOR CAN CAUSE CONTAMINATION OF THE COMPRESSOR.

(2) Apply small quantity of rust inhibitor (PWC15-011) or corrosion preventative compound (PWC15-015) or corrosion inhibitor compound (PWC15-016) or equivalent to all corroded areas on the air inlet case with a spray can or equivalent, with a heat shrink tube (0.13 in. or larger diameter) and 0.024 in. safety wire.

(3) Spray rust inhibitor (PWC15-011) or corrosion preventative compound (PWC15-015) or corrosion inhibitor compound (PWC15-016) or equivalent between the air inlet case and the compressor housing with spray can, 0.13 in. heat shrink tube and 0.024 in. safety wire.

(4) Apply rust inhibitor (PWC15-011) or corrosion preventative compound (PWC15-015) or corrosion inhibitor compound (PWC15-016) or equivalent as a drop between the air inlet case and the No. 1 bearing housing at 12 o'clock position with spray can, large diameter heat shrink tube and 0.024 in. safety wire.

(5) Examine the application of corrosion inhibitor on the air inlet case with a boroscope.

NOTE: 1. The corrosion inhibitor can also be applied on the air inlet case by connecting the heat shrink tube to the borescope with a clip.

NOTE: 2. It is possible to do this procedure through the air inlet plenum access cover and inlet screen.

(6) If there is too much corrosion on the air inlet case surface, do the steps that follow:

(a) Remove the inlet case screen (Ref. 72-20-00, COMPRESSOR AIR INLET CASE - MAINTENANCE PRACTICES).

(b) Soak a lint free cloth in the rust inhibitor (PWC15-011) or corrosion preventative compound (PWC15-015) or corrosion inhibitor compound (PWC15-016) or equivalent, then apply corrosion inhibitor on the air inlet case.

(c) Install the inlet case screen (Ref. 72-20-00, COMPRESSOR AIR INLET CASE - MAINTENANCE PRACTICES).

8. Control Linkage

Inspect all engine controls for correct operation. Make sure the linkage from the cockpit controls to

the engine control levers are connected to permit full and free movement. Make sure controls are adjusted to permit over-travel of power control lever to ensure full operation of engine and that the linkage can be operated without interference. Also, investigate and eliminate any slack in the control system. Lubricate and adjust all controls (Ref. Aircraft Maintenance Manual).

9. Periodic Inspection

NOTE: The following tolerance is established for maintenance scheduling convenience.

Unless otherwise stated, the tolerance for periodic inspections is ten percent (10%), or up to a maximum of 100 hours operating time, whichever is less. The tolerance for scheduled inspections is ten percent (10%), or up to a maximum of 30 days calendar time, whichever is less.

Subsequent intervals will be adjusted to re-establish the original schedule. When an inspection is done more than 10% early, subsequent inspections will be advanced as required to not exceed the maximum tolerance.

For inspection items and intervals, refer to Table 601. Refer to Inspection/Check in the referenced CHAP/SEC/SUB for limits and disposition of items.

Table 601 Periodic Inspection

Component				Inspection	Interval
1.	Engine Externals				
	A.	Tubing, Wiring, Control linkages, Hose Assemblies	(1)	All accessible connections, clamps and brackets for attachment.	MINOR
				NOTE: Inspect accessible lockwire and safety cable for security and installation.	
			(2)	Wear, chafing, cracks and corrosion.	MINOR
				NOTE: Visually inspect insulated air tubes for sign of swelling, cracking, bulging of rubber sheath material. Refer to repair section and SB1687. Replace as necessary.	
			(3)	Fuel and oil lines for leaks.	ROUTINE
			(4)	Depending on operating conditions and environment, examine linkages at regular intervals. Pay particular attention to rear linkage cam box, fuel control unit arm, telescopic rod and rod end fittings.	MINOR (See NOTES)

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Component				Inspection	Interval
				Disconnect rod ends and clean with solvent (PWC11-027) or (PWC11-031). Examine rod end for corrosion, roughness in rotation, side play and radial play and replace if noted. Lubricate with light grease (PWC04-001), reinstall rod ends and torque to specified value (Ref. 76-10-00, PROPELLER REVERSING LINKAGE - MAINTENANCE PRACTICES). Check free movement of linkage.	
				NOTE: 1. Inspect initially at 100 hours with extension to relevant airframe zone inspection based on inspection results but not to exceed 200 hours.	
				NOTE: 2. With the exception of rod end fittings, linkages generally will operate satisfactorily without lubrication. While lubrication will be effective in some instances, it must be realized that grease and oil attracts dirt and foreign matter. Depending on local conditions, operators should take these facts into consideration before deciding to lubricate components.	
	B.	Air Inlet Screen		Cleanliness (Ref. 72-20-00).	MINOR
	C.	Gas Generator Case		External surfaces, and fire seal mount ring brackets for cracks, distortion and corrosion (Ref. 72-30-04)	Minor
	D.	Fireseal Mount Rings		Cracks and attachment of brackets and seals. (Ref. 72-30-01/-02)	MINOR
	E.	Exhaust Duct	(1)	Cracks and distortion (Ref. 72-50-05, MAINTENANCE PRACTICES).	MINOR
			(2)	Engines that exhibit inferior welds (Ref. 72-50-05, MAINTENANCE PRACTICES) (Ref. SB1610)	MINOR not to exceed 150 hours.
			(3)	Engines that exhibit cracks (Ref. 72-50-05, MAINTENANCE PRACTICES)(Ref. SB1610)	25 hours

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Component				Inspection	Interval
	F.	Propeller Shaft Seal		Check for oil leaks (Ref. 72-10-00)	ROUTINE
	G.	Accessories	(1)	Attachment of accesories and linkages, air, oil and fuel lines (Ref. 73-10-07/-08) .	MINOR
				NOTE: Visually inspect insulated air tubes for sign of swelling, cracking, bulging of rubber sheath material. Refer to repair section and SB1687. Replace as necessary.	
			(2)	Security of pneumatic lines (Ref. 73-10-07/-08).	MINOR
			(3)	Oil and fuel leaks as applicable.	ROUTINE
			(4)	The starter-generator gearshaft splines for wear (Ref. 72-60-00). If wear found is above the limits, replace the gearshaft within the next 50 flight hours.	At the starter-generator removal/replacement only.
2.	Engine Internals				
	A.	First-stage Compressor Rotor		Do a visual inspection with a mirror or a borescope (Ref. Para. 10.D.).	MINOR
				NOTE: If any FOD has caused the engine to become prone to engine surges or causes unusual compressor whining, the large bent airfoil will make the whining sound (Ref. Fig. 604). Remove the engine immediately.	
	B.	Compressor Turbine Disk Assembly		Overhaul. Remove assembly and ship to an approved overhaul facility.	Refer to SB1803.
	C.	Hot Section		Examine with borescope (Ref. Para. 10.).	In conjunction with periodic fuel nozzle leak and function tests (Ref. 3.B.(11)).
3.	Systems				
	A.	Oil System	(1)	Check oil level (Ref. SERVICING, Lubricating Oil System).	ROUTINE
				NOTE: For oil change recommendations, refer to **SB1001**.	

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Component				Inspection	Interval
			(2)	Check condition and locking of oil filler cap (Ref. 72-60-00).	ROUTINE
			(3)	Oil filter elements and secondary screen (coarse hat-type screen attached to the inner end of the filter). Elements must be inspected and cleaned (Ref. 79-20-02). Light traces of sediment only may be removed from the main filter screen. All other contamination requires replacement of filter element. Any foreign material found in main oil filter or on chip detector, should be identified before further inspection/operation (Ref. 70-00-00, MAINTENANCE PRACTICES).	MINOR (See NOTE)
				NOTE: 1. Inspect initially at 100 hours with extension to relevant airframe zone inspection based on inspection results but not to exceed 200 hours.	
				NOTE: 2. If the engine has a recent history (within 200 flight hours) of generating carbon debris; do the inspection at 100 hours interval and replace the filter element at 300 hours interval. Refer to 72-00-00, ENGINE, TURBOPROP - FAULT ISOLATION for the instructions to start oil filter inspection again and for the oil filter replacement intervals.	
				NOTE: 3. If carbon like deposits are found, drain accessory gearbox oil into a clean container and examine debris. Varnish flakes are non metallic and are usually dark on one side and shiny, similar to bronze, on the other side. They are hard and will not form into paste when rubbed between fingers (Ref. step (7)).	
			(4)	Use approved equipment to clean permanent filter element at an overhaul facility prior to further use. Do not clean ultrasonically. Filter element must be discarded after 1000 hours or after heavy contamination.	
				NOTE: While the chip detector is removed, inspect with both poles bridged (Ref. Step 5). With poles unbridged, inspect for continuity. If continuity exists, replace the chip detector.	

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Component				Inspection	Interval
			(5)	Check magnetic chip detector(s) for continuity, open circuit must exist indicating no contamination at pole tips. If continuity is there, remove and examine for contamination. Any foreign material found on the chip detector or in the main oil filter must be identified before further inspection/operation (Ref. 79-20-02, OIL FILTER, HOUSING AND CHECK VALVE - MAINTENANCE PRACTICES) (See NOTE 3).	Every 100 hours (See NOTES 1 and 2)
				NOTE: 1. FOR SINGLE ENGINE AIRCRAFT, If the chip detector is connected to a cockpit indicating system, examine the chip detector initially at 100 hours with extension to relevant airframe zone inspection based on the inspection results, but not more then 200 hours.	
				NOTE: 2. FOR TWIN ENGINE AIRCRAFT, If the chip detector is connected to a cockpit indicating system, examine the chip detector initially at 100 hours with extension to relevant airframe zone inspection based on the inspection results, but not more than 400 hours.	
				NOTE: 3. When you remove the chip detector, examine with both poles bridged (Ref. Step (6)). Do an inspection for continuity with poles unbridged. If there is continuity, replace the chip detector.	
			(6)	Remove the magnetic chip detector (Ref. 72-10-00). Bridge the chip detector's magnetic poles with an applicable jumper	Every 600 hours or one year, whichever comes first.

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Component				Inspection	Interval
				and check for continuity between the connector pins. Replace the chip detector if there is no continuity. Any foreign material found on the chip detector or in the main oil filter housing must be identified and categorized before further inspection/operation (Ref. 79-20-02, OIL FILTER, HOUSING AND CHECK VALVE - MAINTENANCE PRACTICES).	
			(7)	Check scavenge oil pump housing for leaks.	ROUTINE
			(8)	Check the AGB internal scavenge oil pump inlet screen (Ref. Chapter 72-60-00 CLEANING/INSPECTION). Collect drained oil. Using a mirror and a flashlight inspect the scavenge oil pump inlet screen. Any foreign material found blocking the screen or contained in the oil should be identified before further operation (Ref. Unscheduled Inspection).	200 hours or 6 months (at operator's option) for engines operating in high relative humidity/tropical environment (above 70% for most of the year).
					Extension in steps of 50 hours to 500 hours maximum, based on operator experience.
					1000 hours for engines NOT operating in high relative humidity/tropical environment.
	B.	Fuel System	(1)	Check the fuel pump (in-situ inspection only) for installation and leaks (Ref. 73-10-02, FUEL PUMP - MAINTENANCE PRACTICES).	MINOR
			NOTE: If airframe fuel boost pump fails or is inadvertently left off for an accumulative time in excess of 10 hours, the engine driven fuel pump must be removed and replaced. The removed pump should be sent to an approved overhaul facility.		
			(2)	Check inlet screen for foreign matter or	Every 600 hours

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Component	Inspection	Interval
	distortion, clean and reinstall, or install new screen. (Ref. 73-10-02).	
	CAUTION: ON NEW AIRCRAFT, CHECK FILTER AFTER EACH FLIGHT UNTIL NO CONTAMINATION IS EVIDENT. CHECK AFTER FIRST FLIGHT OR GROUND RUN WHENEVER ANY COMPONENT UPSTREAM OF FILTER IS REPLACED.	
(3)	Check outlet filter for foreign matter or distortion (Ref. 73-10-02). Install new filter as service conditions dictate, not to exceed 600 hours and when fuel system contamination is suspected.	MINOR
(4)	If Sundstrand fuel pump is installed:	
	If fuel pump gear-set and coupling operating time since new (TSN) is **LESS THAN** Engine Basic TBO (Ref. **SB1803**):	
	(a) Check fuel pump coupling in-situ for fretting and corrosion (Ref 73-10-02, FUEL PUMP - MAINTENANCE PRACTICES).	Every 600 hours
	(b) Remove fuel pump and inspect the drive coupling and cover accessory gearbox side for signs of reddish-brown (iron oxide) stains. If stains are observed, return the fuel pump to an approved overhaul facility (Ref. 73-10-02, FUEL PUMP - MAINTENANCE PRACTICES).	Every 1800 hours
	If fuel pump gear-set or coupling operating time since new (TSN) is **MORE THAN** Engine Basic TBO (Ref. **SB1803**) **OR** is **UNKNOWN OR** fuel pump was overhauled before 2007:	
	(a) Remove fuel pump and inspect the drive coupling and cover accessory gearbox side for signs of reddish-brown (iron oxide) stains. If stains are observed, return the fuel pump to an approved overhaul facility (Ref. 73-10-02, FUEL PUMP - MAINTENANCE PRACTICES).	300 hours

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Component			Inspection		Interval
				NOTE: As of January 2010, the fuel pump gear-set and drive coupling replacement is required at every pump overhaul.	
			(5)	Check the drain valves (in-situ inspection only) for installation and leaks (Ref. 73-10-06, FUEL DRAIN VALVES - MAINTENANCE PRACTICES).	MINOR
			(6)	Check the flow divider, dump valve, and starting control (in-situ inspection only) for installation and leaks (Ref. 73-10-04, FLOW DIVIDER AND STARTING CONTROL - MAINTENANCE PRACTICES).	MINOR
			(7)	Check FCU for installation, linkages and pneumatic tubes (Ref. 73-20-00). Evidence of FCU bearing washout indicated by traces of blue dye effluent is caused by a mixture of bearing grease and fuel.	MINOR
			(8)	For **Post-SB1568**: Do a deceleration check (Ref. 71-00-00, POWER PLANT - ADJUSTMENT/TEST).	150 hours or MINOR, whichever occurs first for single engine installation or 400 hours for twin engine installation.
				NOTE: Not required for FCU identified with "RE52", "SB 73-3" or with a serial number that contains the letter F as prefix.	
			(9)	For Post-SB1472 engines fitted with a manual override on the fuel control, check FCU Manual Override System for static operation (Ref. 71-00-00, ADJUSTMENT/TEST).	MINOR
			(10)	(a) Remove the FCU (Ref. 73-20-00, FUEL CONTROL UNIT - MAINTENANCE PRACTICES) for drive body inspection/driveshaft bearing replacement in accordance with the	

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Component	Inspection	Interval
	component maintenance manual.	
	Engine operating frequently or continually in salt-laden (coastal, island and sea surveillance) or severe to extreme polluted or sandy/dusty environments.	1800 hrs or 6 years, whichever comes first (Ref. NOTES 1, 2, 4 and 5)
	Engine operating occasionally or rarely in salt-laden (coastal, island and sea surveillance) or severe to extreme polluted or sandy/dusty environments.	12 years (Ref. NOTES 3, 4 and 5)
	NOTE: 1. For maintenance scheduling purposes, the FCU may be operated for an additional 6 months or 200 flight hours, whichever comes first. This may also be used as a one-off scheduling allowance for the first inspection from 31 July 2023.	
	NOTE: 2. If the FCU has exceeded 1800 flight hours or 6 years since new or last FCU overhaul or last drive body inspection, the operator may schedule the FCU shop visit within the following calendar transition period (from 31 July 2023): (a) If the interval was exceeded after 31 July 2017 (within 6 years), P&WC recommend to comply within 24 months. (b) If the interval was exceeded prior to 31 July 2017 and after 31 July 2011 (between 6 to 12 years), P&WC recommend to comply within 18 months. (c) If the interval was exceeded prior to 31 July 2011 (more than 12 years), P&WC recommend to comply within 12 months.	
	NOTE: 3. If the FCU has exceeded 12 years since new or since last overhaul or drive body inspection, the operator may schedule the FCU shop visit within the following calendar transition period (from 31 July 2023): (a) If the interval is less than 22 years (last performed after July 2001), P&WC recommend to comply within 36 months (prior to July 2026). (b) If the interval is greater than 22 years (last performed prior to July 2001), P&WC recommend to comply within 12 months (prior to July 2024).	

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Component				Inspection	Interval
				NOTE: 4. To determine the operating frequency in harsh environments, refer guidance in Para. 5. or Chapter 71-00-00, POWER PLANT - CLEANING.	
				NOTE: 5. For engines operating to a P&WC recommended TBO escalation program, based on operator experience and reject rate (with the FCU) and no significant change in the environment or mission since issue of the P&WC recommended TBO escalation, the FCU drive body inspection/driveshaft replacement can be aligned with the P&WC recommended engine TBO interval.	
				(b) Remove FCU and send for overhaul.	Ref. SB1803.
			(11)	Leak test and function test fuel manifold adapter and nozzle assemblies. Clean as necessary (Ref. 73-10-05).	See NOTES
				NOTE: 1. Do the periodic borescope inspection of the hot section components in conjunction with these tests (Ref. 2.C.).	
				NOTE: 2. If one or more unacceptable nozzle(s) is (are) found during the leak and function tests, it is highly recommended that the hot section be inspected for damage by a direct visual inspection or borescope inspection.	
				NOTE: 3. Regular fuel nozzle maintenance is important for hot section durability. Inspection is recommended at routine intervals according to either of the following:	
				NOTE: 4. Engines **ON** fuel nozzle in-situ cleaning program (Ref.71-00-00, POWER PLANT - CLEANING), test fuel nozzles and repair as necessary (Ref. 73-10-05, FUEL MANIFOLD AND NOZZLES - MAINTENANCE PRACTICES). Where fuel quality is questionable, and for operators new to PT6A operation, inspection is recommended at 400 hours initially, with extension based on the operators experience and rejection rate, not more than 200 hours. For other operators, inspection is recommended at 600 hours initially, with extension based on the operators experience and rejection rate, not more than 200 hours.	
				NOTE: 5. Engine **NOT ON** fuel nozzle in-situ cleaning program, test	

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Component				Inspection	Interval
				fuel manifold adapter (Ref. 73-10-05, FUEL MANIFOLD AND NOZZLES - MAINTENANCE PRACTICES). Where fuel quality is questionable, and for operators new to PT6A operation, cleaning/inspection is recommended at 200 hours initially, with extension, based on the operators experience and rejection rate, not more than 200 hours. For other operators, inspection is recommended at 400 hours initially, with extension based on the operators experience and rejection rate, not more than 200 hours.	
			(12)	Check the oil-to-fuel heater (in-situ inspection only) for installation (Ref. 73-10-01, OIL-TO-FUEL-HEATER - MAINTENANCE PRACTICES).	MINOR
	C.	Ignition System	(1)	Examine the ignition exciter/current regulator for installation and condition (Ref. 74-10-01, IGNITION CURRENT REGULATOR - MAINTENANCE PRACTICES).	400 Hours
			(2)	Examine the ignition cables for chafing, wear and installation (Ref. 74-20-01, IGNITION CABLES - MAINTENANCE PRACTICES).	400 Hours
			(3)	Examine the spark igniters/glow plugs for cleanliness and erosion. Do a functional check (Ref. 74-20-02, GLOW PLUGS - MAINTENANCE PRACTICES and Ref. 74-00-00, IGNITION - MAINTENANCE PRACTICES).	400 Hours
	D.	Pneumatic System	(1)	Check P3 filter for installation (Ref. 73-10-07).	MINOR
			(2)	Inspect pneumatic tubes and lines for cracks and damages especially at the end fittings and joints.	MINOR
			(3)	Clean or replace filter, dependent on condition, service experience or environment (Ref. 73-10-07).	See NOTE 2
				NOTE: Visually inspect the P3 filter for corrosion. If the filter shows presence of corrosion, discard	

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Component			Inspection		Interval
				the filter and replace with a new filter.	
			(4)	Clean and inspect Post-SB1495 P3 filter drain valve housing assembly (Ref. 73-10-07).	See NOTE 2
			NOTE: 1. Filter contamination is greatly influenced by particular operating conditions and environment, inspection intervals may be modified accordingly.		
			NOTE: 2. Inspect initially at 100 hours with extension to relevant airframe zone inspection based on inspection results, but not to exceed 200 hours.		
			(5)	Replace disposable filter based on condition, service experience or environment.	Every 1000 hours.
			(6)	Clean or replace permanent filter based on condition, service experience or environment. Ship to an approved overhaul facility for ultrasonic cleaning.	Every 1000 hours.
			NOTE: Send filter to an approved facility for ultrasonic cleaning and testing in accordance with the engine overhaul manual every 1000 hours.		
			(7)	Check the bleed valve (Ref. 75-30-00, Inspection/Check).	
				(a) For engines operating in salt laden or harsh environments or high cyclic conditions.	Every 600 hours. Extension can be based on operators experience.
				(b) For other engines NOT operating in conditions in Step (7)(a).	Inspect in conjunction with a Hot Section Inspection.

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Figure 604 Large Bent Airfoil on First-Stage Compressor Rotor



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10. Borescope

NOTE: The borescope inspection is not intended to replace or diminish the importance of doing an HSI, but is intended to monitor the internal components of the engine to help reduce the cost of future repairs/refurbishments.

A. General

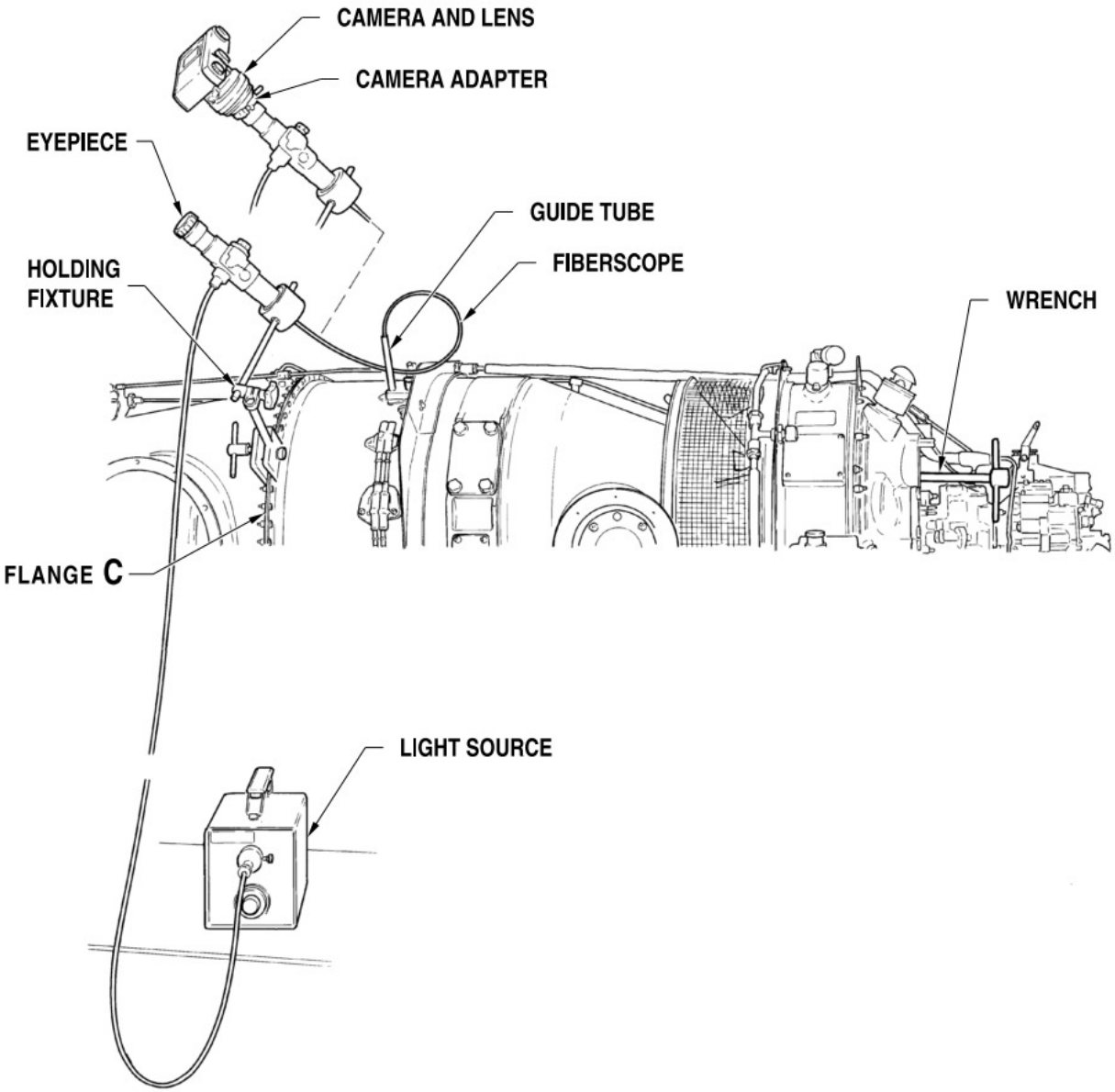
CAUTION: HEAT CAN DAMAGE THE BORESCOPE. ENGINE TEMPERATURE MUST BE LESS THAN 66°C (150°F) BEFORE YOU DO AN INSPECTION. THE SATISFACTORY COOLING PERIOD FOR STANDARD AMBIENT CONDITIONS IS 40 MINUTES (COLD SECTION) OR TWO HOURS (HOT SECTION) AFTER ENGINE SHUTDOWN, BUT CAN CHANGE WITH AMBIENT CONDITIONS. IF REQUIRED, DO DRY MOTORING RUNS TO ACCELERATE COOLING (REF. 71-00-00, POWER PLANT - ADJUSTMENT/TEST).

The borescope is a optical fibrescope or a digital devise which lets an operator to inspect hot section areas of the engine without removal or disassembly of the engine. Access is through ports or openings created by removal of engine components. Personnel that do the inspection should be qualified to use the borescope and analyze results. The borescope is a delicate device vulnerable to shocks, twisting and pinching. Use care and attention when handling tool. Examine the borescope assembly, it's accessories (Ref. Fig. 605), and the following procedures.

NOTE: As an alternative, you can remove the Power Section Module (Heavy Maintenance only) (Ref. Para. 11. B.).

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Figure 605 Borescope and Accessories Installed



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B. Description

The borescope assembly comprises a pattern-controlled rigid guide tube, a direct viewing flexible 5 mm fibrescope, a light source, a side viewing adapter, and other accessories. If required, a camera may be used to record engine areas being inspected.

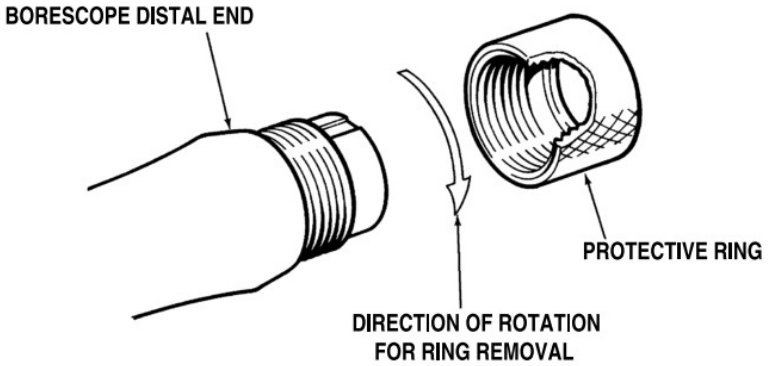
An operator can do on-wing inspections of the CT stator assembly, combustion chamber liners, including cooling rings and dome, and AGB and RGB gears.

C. Removal/Installation of Borescope and Accessories

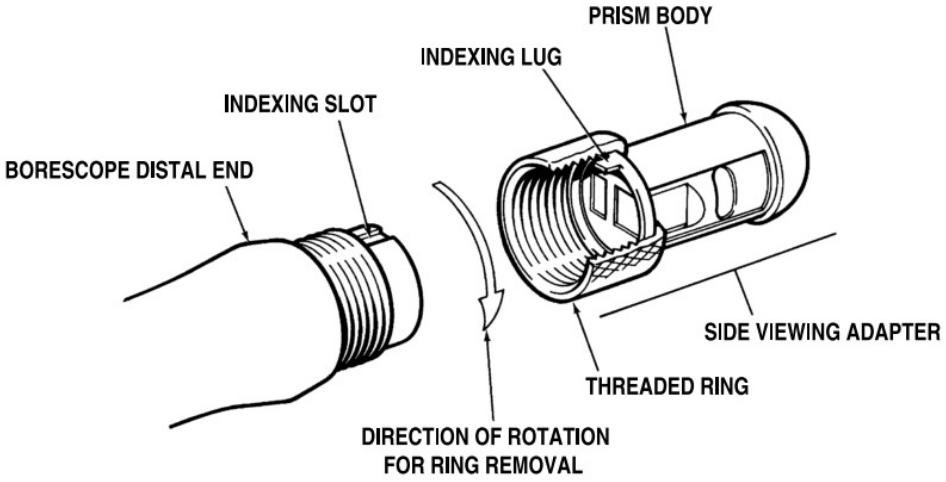
(1) Side viewing adapter (Ref. Fig. 606).

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Figure 606 Removal/Installation of Side Viewing Adapter



RING REMOVAL



ADAPTER INSTALLATION

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NOTE: An adapter (SL-4TP) is used to inspect components located at a nominal 90 degree angle to fiberscope distal tip. The use of side viewing adapter will introduce a reduction in the field of view as compared to the direct viewing field. A ring is installed to protect distal end when side viewing adapter is not fitted.

(a) Installation

1 Hold fiberscope as close as possible to distal end, and remove protective ring.

CAUTION: INSTALL SIDE VIEWING ADAPTER CAREFULLY. IF NOT INSTALLED AND TIGHTENED CORRECTLY, ADAPTER COULD FALL INTO ENGINE. OVERTIGHTENING COULD DAMAGE DISTAL END.

2 Hold fiberscope as close as possible to distal end, and install adapter with indexing slot and lug aligned. Tighten adapter fingertight.

(b) Removal

1 Hold fiberscope as close as possible to distal, and remove side viewing adapter.

CAUTION: INSTALL PROTECTIVE RING CAREFULLY. IF NOT INSTALLED AND TIGHTENED CORRECTLY, RING COULD FALL INTO ENGINE. OVERTIGHTENING RING COULD DAMAGE DISTAL END.

2 Hold fiberscope as close as possible to distal end and install protective ring. Tighten ring fingertight.

(2) Light source

NOTE: Specify power requirement when purchasing borescope.

(a) A halogen lamp, with either 110 V 60 Hz, or a 220V 50 Hz power supply.

(b) Remove top cover from light source to replace lamp.

(c) Set intensity knob at maximum for best results.

(d) Before installing light source, refer to manufacturer's instructions.

(3) Camera

(a) A camera may be used to record internal engine components. It must be equipped with a 50mm F1.8 lens (Ref. Camera Handbook).

(b) Installation

CAUTION: DO NOT USE COMPRESSED AIR TO CLEAN CAMERA, BORESCOPE, OR ASSOCIATED EQUIPMENT

1 Clean camera viewfinder, reflex mirror, focusing screen, and lens with lens cleaning tissue and cleaner.

2 Install endoscopic focusing screen in camera.

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3 Install lens.

4 Install adapter (CAII) on lens.

5 Install film in camera (Ref. Camera Handbook).

6 Set camera film speed to suit film, and set exposure to -2 (Ref. Camera Handbook).

7 Align bayonet slots of adapter and install camera on borescope eyepiece pins.

NOTE: Make sure distal end is not moved while taking photographs.

(c) Removal

1 Turn outer ring of camera adapter to disengage locking mechanism and remove camera from borescope eyepiece.

2 Remove camera adapter from lens.

3 Remove film from camera and add following data:

Engine S/N

Date and area photographed

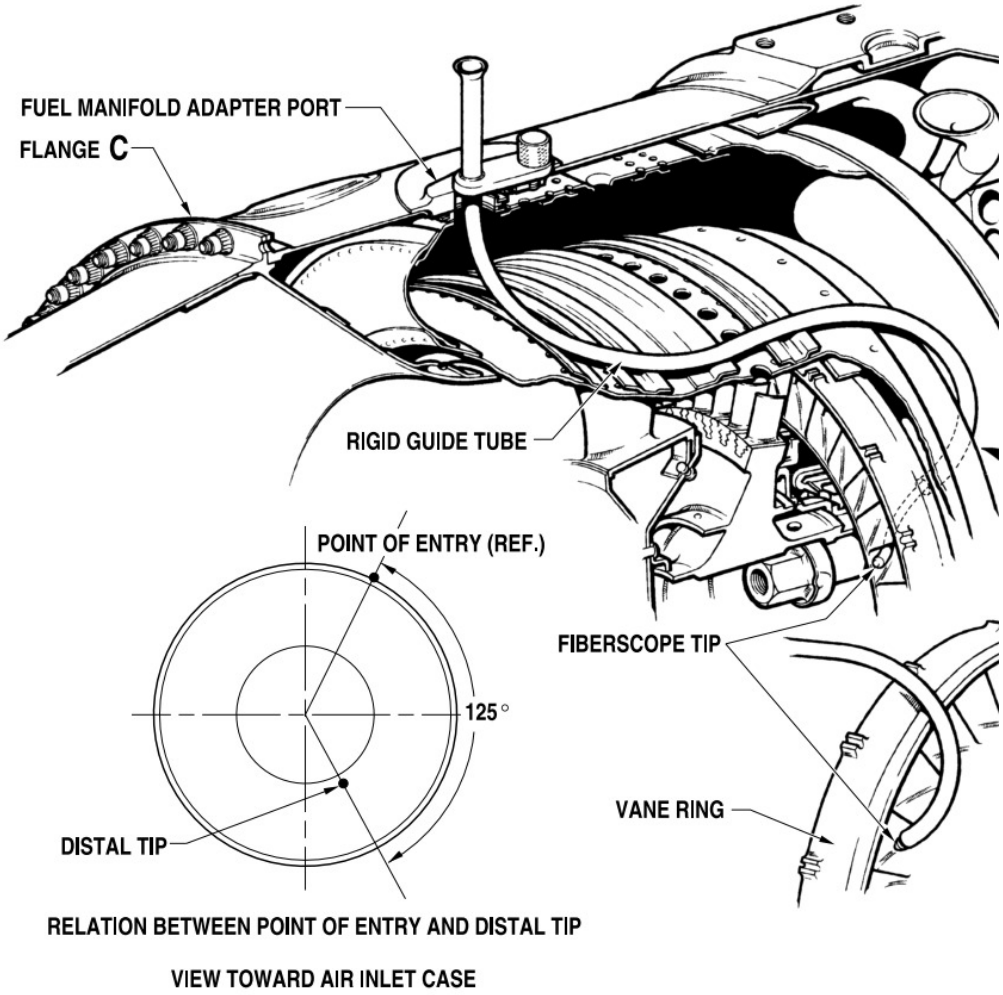
Engine TSN/TSO or cycles since last overhaul

Reason for borescope inspection (suspected foreign object damage, low power, etc.)

(4) Guide Tube (Ref. Fig. 607)

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Figure 607 Guide Tube Orientation



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(a) Installation

1 Remove appropriate fuel manifold adapter(s) (Ref. 73-10-05).

2 Ease the guide tube (PWC34910-200) through the manifold adapter opening into the combustion chamber and exit duct area turning the guide tube three-quarters of a turn CCW.

NOTE: The rigid, pattern controlled guide tube must be inserted freely with NO force.

3 The guide tube end locates between vane of CT stator, while the supporting flange is attached to the adapter boss.

(b) Removal

1 Loosen knurled screw to release guide tube supporting flange.

2 Withdraw guide tube, turning clockwise.

3 Install fuel manifold adapter(s) (Ref. 73-10-05).

D. Inspection

(1) Combustion Chamber Liner Assembly and Large Exit Duct

CAUTION: MAKE SURE ENGINE TEMPERATURE IS BELOW 60°C (140°F).

(a) Remove fuel manifold adapter as necessary (Ref. 73-10-05).

(b) Fasten holding fixture (PWC34913) to Flange C, insert fiberscope into fuel manifold adapter port, fasten eyepiece to fixture and connect light source.

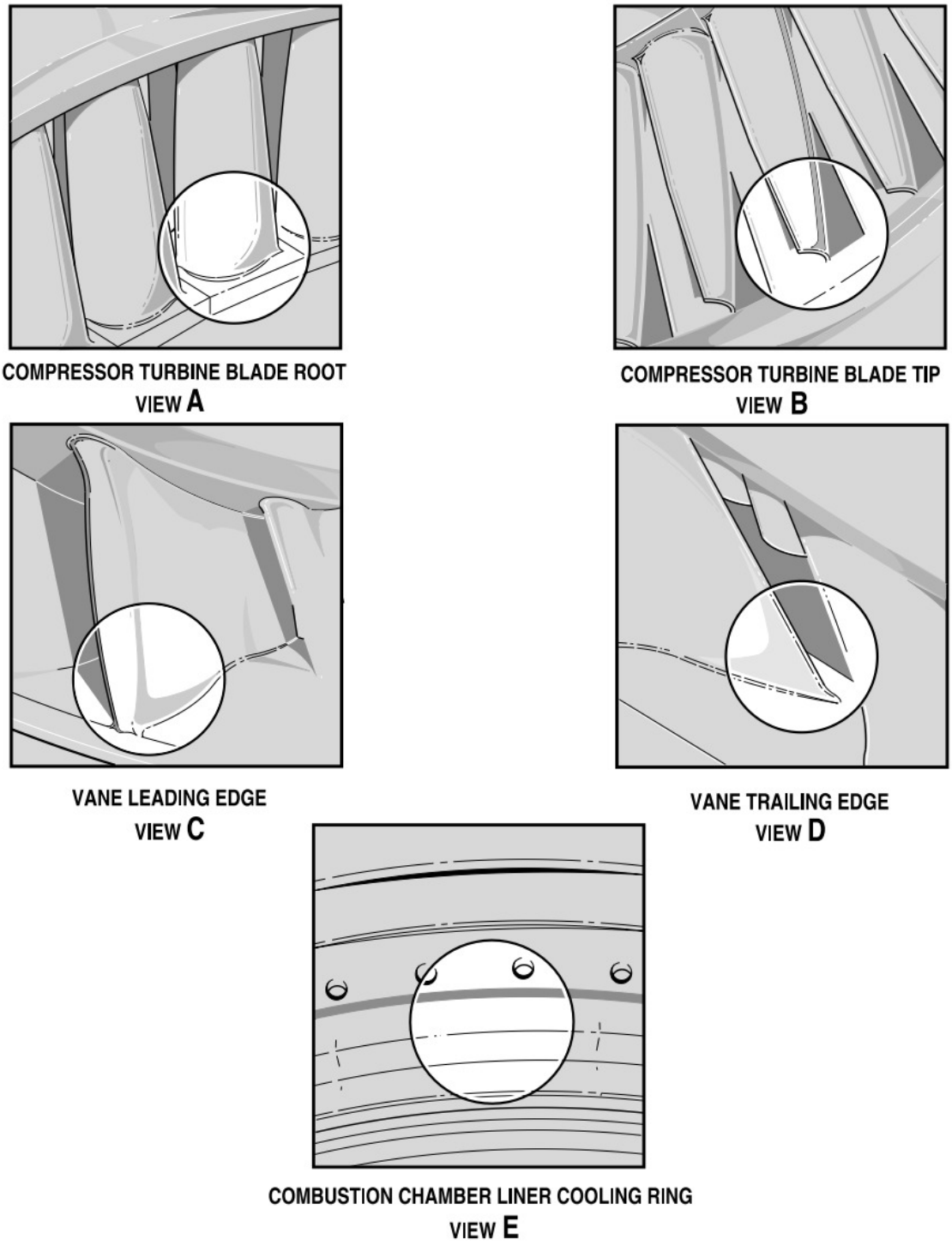
(c) Examine combustion chamber liner (Ref. Para. 11.C., Fig. 608, View E and Fig. 609, Area 1 and Area 2).

(d) Remove fiberscope, light source and holding fixture.

(e) Install fuel manifold adapter (Ref. 73-10-05).

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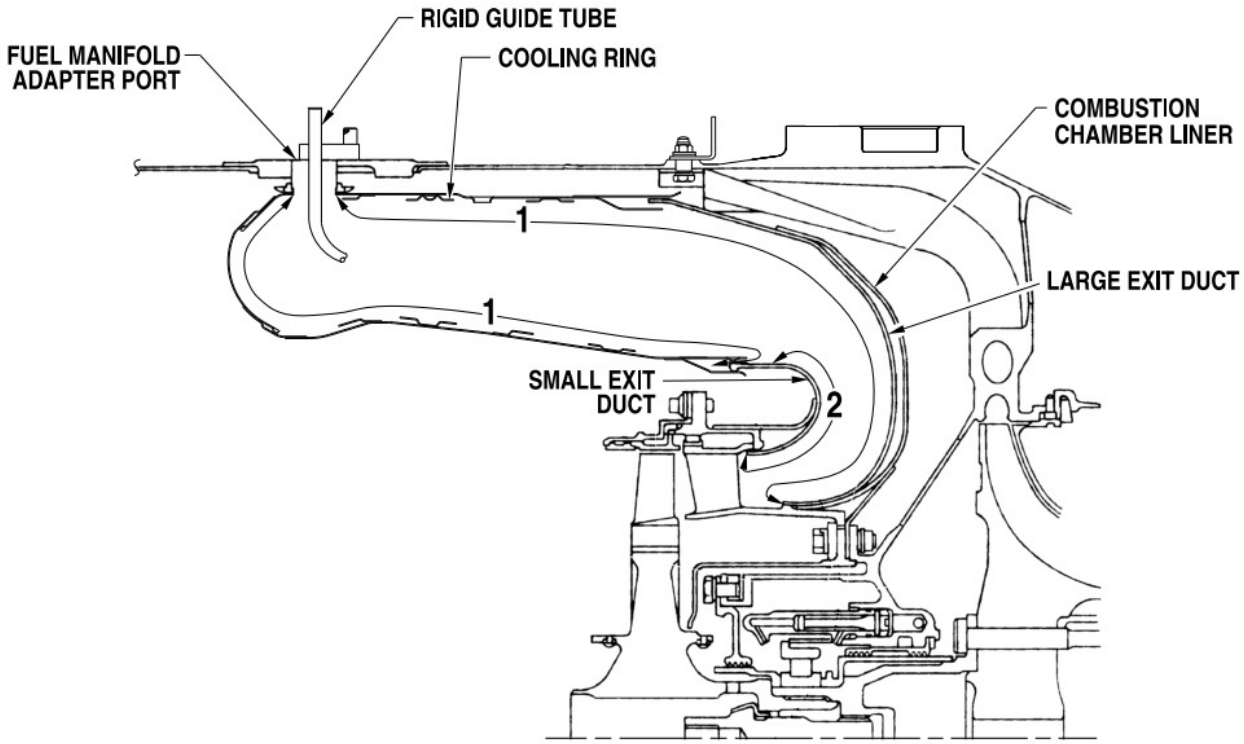
Figure 608 Borescope Views



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Figure 609 Inspection of Combustion Chamber Liner Assembly and Large Exit Duct



NOTES:

AREA 1: COMBUSTION CHAMBER INNER/OUTER LINER AND LARGE EXIT DUCT.
AREA 2: SMALL EXIT DUCT.

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(2) CT Stator Assembly

(a) Install guide tube (Ref. Para. C. step (4)).

(b) Fasten holding fixture (PWC34913) to Flange C, install borescope, and attach light source (Ref. Fig. 605).

CAUTION: MAKE SURE ENGINE TEMPERATURE IS BELOW 60°C (140°F).

(c) Insert fiberscope into guide tube.

NOTE: The side viewing adapter will add to inspection capability (Ref. Para. C. step (1)).

(d) Examine the CT stator assembly (Ref. Para. 11. D.).

(e) Remove light source, borescope and holding fixture.

(f) Remove guide tube (Ref. Para. C. step (4)).

(3) CT blades and shroud segments

(a) Install guide tube (Ref. Para. C. step (4)).

(b) Fasten holding fixture (PWC34913) to Flange C (Ref. Fig. 605).

(c) Install borescope on holding fixture and connect light source.

CAUTION: MAKE SURE ENGINE TEMPERATURE IS BELOW 60°C (140°F).

(d) Insert fiberscope into guide tube while looking through eyepiece. Stop inserting when distal tip reaches end of guide tube.

NOTE: Make sure distal tip actuating lever is in neutral position when installing/removing fiberscope.

(e) Remove starter-generator (Ref. Aircraft Maintenance Manual).

(f) Install wrench (PWC34941) on AGB splined gearshaft.

(g) Loosen knurled knob on holding fixture.

CAUTION: MAKE SURE THAT DISTAL TIP IS NOT BETWEEN THE CT BLADES BEFORE TURNING THE COMPRESSOR.

(h) Examine the CT blades and shroud segments (Ref. Paras. 11. E. and F.), while a second operator rotates the compressor use the wrench.

(i) Retract fiberscope into guide tube.

(j) Remove fiberscope from guide tube.

(k) Remove borescope and holding fixture.

(l) Remove guide tube (Ref. Para. C. step (4)).

(m) Insert fiberscope through the exhaust duct port.

NOTE: 1. Regular borescope inspections of the turbine blades help the operator to

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make sure if the wash/rinse schedule is sufficient to prevent corrosion. Operators experience can necessitate a more aggressive wash schedule for optimal performance and reduced maintenance costs (Ref. 71-00-00, POWER PLANT - CLEANING).

NOTE: 2. Based on the operator experience, examine the PT blades for sulphidation.

CAUTION: MAKE SURE THE PROPELLER AND THE POWER TURBINE DO NOT ROTATE WHILE FIBRESCOPE IS INSERTED THROUGH THE PT BLADES.

(n) Use distal tip actuating lever and carefully insert fiberscope or videoscope through the PT blades and the PT vane ring.

(o) Examine the trailing edge of the CT blades (Ref. Para. 11. E.), while a second operator rotates compressor use the wrench.

(p) Carefully remove the fiberscope from the exhaust duct.

(q) Remove wrench from splined gearshaft, and install starter generator (Ref. Aircraft Maintenance Manual).

(4) First-stage Compressor Rotor and Air Inlet Surfaces

CAUTION: KEEP COMPRESSOR INLET AREA CLEAR OF ALL FOREIGN OBJECTS. FOREIGN OBJECTS CAN MOVE TO DOWN STREAM ENGINE COMPONENTS AND CAUSE DAMAGE DURING SUBSEQUENT OPERATION.

(a) Install the holding fixture (PWC34913) to the engine rear lifting bracket, and attach the eyepiece to the holding fixture, and connect light source.

CAUTION: DO NOT TRY TO PUSH THE FIBERSCOPE MORE THAN THE FIRST-STAGE COMPRESSOR STATOR.

(b) Put the fiberscope into the compressor inlet for the inspection of the first-stage compressor rotor and the air inlet case surfaces.

(c) Examine the first-stage compressor rotor (Ref. 72-30-05, COMPRESSOR ROTOR - MAINTENANCE PRACTICES).

(d) Examine the compressor air inlet surfaces for corrosion (Ref. 72-20-00, COMPRESSOR AIR INLET CASE - MAINTENANCE PRACTICES).

(e) Remove the fiberscope, light source, eyepiece and the holding fixture (PWC34913).

(5) Accessory Gearbox Gears

(a) Remove oil filler cap/ gage assembly and filler tube from accessory gearbox housing (Ref. 72-60-00).

(b) Remove starter-generator (Ref. AMM 80-10-00).

NOTE: When the AGB inspection is done at the same time as the first stage compressor rotor inspection, the AGB gears can be turned by rotating the first stage rotor. Do not use bare hands.

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(c) Insert wrench (PWC34941) into splined gearshaft.

(d) Clamp holding fixture (PWC34913) to engine rear lifting bracket, fasten eyepiece to holding fixture, and connect light source.

CAUTION: MAKE SURE THAT END OF FIBERSCOPE DOES NOT TOUCH THE GEARS.

(e) Put the fiberscope into the borescope port and examine the gear teeth (Ref. Para. 11. I.), while a second operator rotates the geartrain use the wrench.

(f) Inspect gearbox internally for corrosion.

(g) Remove wrench, light source, fiberscope, eyepiece and holding fixture.

(h) Install starter-generator (Ref. AMM 80-10-00).

(i) Install oil filler tube, filler cap and gage assembly (Ref. 72-60-00).

(6) Reduction Gearbox

(a) Remove scavenge oil strainer assembly (Ref. 72-10-00).

(b) Fasten holding fixture (PWC34913) to Flange A, attach eyepiece to fixture, and connect light source.

CAUTION: MAKE SURE THAT END OF FIBERSCOPE DOES NOT TOUCH THE GEARS.

(c) Put the fiberscope into the oil strainer access along the lower passage until the first stage planet gears can be seen. To view the second stage gears, put the fiberscope in an upward direction. Examine the gears (Ref. Para. 11. I.).

(d) Manipulate fiberscope while second operator rotates geartrain by turning the propeller by hand.

(e) Inspect gearbox casing internally for corrosion. Pay particular attention to flange area at six o'clock position.

(f) Remove wrench, light source, fiberscope, eyepiece and holding fixture.

(g) Install scavenge oil strainer assembly (Ref. 72-10-00).

E. Fault Isolation

(1) The possible sources of, and remedies for problems encountered when using borescope are shown in Table 602.

Table 602 Fault Isolation

PROBLEM	SOURCE	REMEDY
Poor illumination	Oil or dirt on distal tip, or side viewing adapter prism	Clean using lens cleaner and tissue.
	Light source intensity	Set switch to HIGH.

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PROBLEM	SOURCE switch set at LOW	REMEDY
	Defective lamp	Replace lamp
	Damaged borescope light tube	Return to manufacturer for repair.
	Defective transformer	Return to manufacturer for repair.
Poor definition	Poor illumination	Return to manufacturer for repair.
	Diopter ring not adjusted correctly	Adjust to suit eyes.
	Damaged fibers in fiberscope, (seen as black dots through viewer)	Return to manufacturer for repair.
Fiberscope distal end does not move when control knob is turned	Damaged control wires	Return to manufacturer for repair.
NOTE: Repairs should be done only by the manufacturer.		

11. In-service Inspection

A. Hot Section Components

In-service inspection is used to determine performance deterioration. Parts can be inspected using a borescope (Ref. Para. 10.) or by removing the power section (Ref. Subpara. B.) following. Refer to FAULT ISOLATION for areas to be inspected.

Using a borescope is more practical for this inspection. Removal and installation of the power section, components/assemblies may result in damage to parts and/or distortion of sealing surfaces causing gas leaks and performance loss.

Acceptable damage must be documented and progression monitored. Record engine/module TSN/TSO and cycles, fuel nozzle ports used, if found by borescope, component, description, location and estimation of damage dimensions/crack length.

If rotating components are found with unacceptable damage during inspection, an HSI must be done before next flight. If damaged non-rotating components are found, except when holes are burnt through the combustion chamber or the CT stator airfoil trailing edge defects are beyond limits (Ref. Inspection), the HSI may be delayed, providing an engine performance/ground power check is done.

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An additional engine performance/ground power check and inspection of the affected area must be done after 50 flight hours. Inspect affected area again in 100 flight hours. If there is no increase in unacceptable damage, go back to the normal inspection intervals as per applicable Maintenance Manual section. If increase in unacceptable damage is visible, subsequent inspections and engine performance/ground power check intervals will depend on the rate of unacceptable damage increase and level of deterioration. Contact the field support representative or customer help desk for more information. Use of an engine in service after components have deteriorated may increase the cost of repairs/refurbishments in the future.

Gas path components downstream of components having material missing must be inspected for secondary damage. Pay special attention to rotating components. CT stator vanes burned through at trailing edge (pressure and suction sides) may mean replacement of the complete set of CT blades, depending on the area of the surface burned through.

The ability of an engine to produce the TO power required by the flight manual for the actual Tam/PA is the only engine airworthiness requirement. However, on some installations the aircraft maintenance manual provides an engine ground power test. This test may be used to confirm TO power is produced within operating limits over the complete range of Tam and PA for which the engine/aircraft is certified. If, at an anticipated high ambient temperature and/or altitude, T5 and Ng will approach or exceed the maximum limits, an HSI is recommended before the actual conditions occur. When T5 and Ng approach, or are anticipated to approach or exceed the maximum limits, troubleshoot the engine/installation before doing an HSI.

Carbon accumulation inside fuel nozzle passages is the principle cause of spray pattern degradation resulting in non-uniform combustion and local high temperature peaks. Exposure to these peaks contributes to premature hot section deterioration. Carbon accumulation is progressive and can affect all nozzles. Therefore, inspect all nozzles (Ref. 73-10-05, Inspection/Check), to minimize premature deterioration occurring at other locations.

If component deterioration exceeds limits, either replace the individual component or do an HSI, depending on the general condition of the hot section and engine performance.

The inspections are recommended concurrent with scheduled engine maintenance checks applicable to individual aircraft installations.

B. Removal/Installation of Power Section (Heavy Maintenance Only)

NOTE: This is an alternative to using borescope.

(1) Removal

(a) Remove power section (Ref. SERVICING).

(b) Remove CT disk assembly (Ref. 72-50-02, Removal/Installation).

(c) Remove combustion chamber inner liner (Ref. 72-40-00, Removal/Installation).

(d) Remove CT stator assembly, shroud housing and small exit duct (Ref. 72-50-01, Removal/Installation).

NOTE: Do not separate parts unless CT stator damage exceeds limits.

(e) Inspect hot section components (Ref. Para. C., D., E., F., G. and H. following).

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(2) Installation

(a) Install CT stator assembly, shroud housing and small exit duct (Ref. 72-50-01, Removal/Installation).

(b) Install CT disk assembly (Ref. 72-50-02, Removal/Installation).

(c) Install combustion chamber inner liner (Ref. 72-40-01, Removal/Installation).

(d) Install power section (Ref. SERVICING).

C. Inspection of Combustion Chamber

NOTE: 1. Combustion chamber includes small exit duct, large exit duct, inner and outer combustion chamber liner assemblies.

NOTE: 2. The Large Exit Duct (LED) is a separated part from the combustion chamber liner. Combustion chamber liners inspection criteria are applicable to LED portion (Ref. Fig. 609, Area 1).

(1) Combustion chamber liner cracks may be repairable, and an HSI is recommended before component replacement is required. Converging cracks in the inner and outer combustion chamber liner walls which do not meet are acceptable. Inspect affected area at 100 hours. Subsequent inspections must not exceed 400 hours.

(2) Plasma top coating (ceramic) loss revealing undercoat (diffused aluminide) on outer and inner liners is acceptable, providing base metal is not burnt or eroded.

(3) Small areas (approximately 1 sq.in.) bulging and/or hot spots on inner or outer liner walls may be repairable and an HSI is recommended. Bulging and/or burning in the dome area, associated or not with axial cracks, are acceptable provided the axial cracks (circumferential cracks are unacceptable) do not exceed 1.0 inch in length or 0.030 inch in width. Engine may remain in service, providing the associated fuel nozzle(s) is/are inspected (Ref. 73-10-05, Inspection/Check) and replaced if not within limits. Subsequent inspections are at operators discretion, but must not exceed 400 hours.

(4) Holes in the inner and outer liner walls are unacceptable. An HSI must be done to replace or repair the affected component, and the associated fuel nozzle(s) inspected (Ref. 73-10-05, Inspection/Check) and replaced if not within limits.

(5) Cracked or distorted cooling rings on liners may be repairable. An HSI is recommended before cooling rings are burned through and pieces enter the gas stream. Engine may remain in service if it is understood that cooling air flow is changed, and the rate at which the combustion chamber deteriorates may increase. Therefore, if an HSI is not done, combustion chamber liner with distorted cooling rings or converging cracks must have the associated fuel nozzle(s) inspected (Ref. 73-10-05, Inspection/Check) and replaced if not within limits. Inspect damage at 100 hours. Subsequent inspections are at operators discretion, but must not exceed 400 hours.

(6) Excessive carbon deposits inside combustion chamber could be result of poor fuel atomization by the fuel nozzles (indicated by deposits around fuel nozzle bosses). Distortion of combustion chamber liner cooling rings may produce carbon deposits in the dome area downstream of the affected cooling ring. If excessive carbon deposits are

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found, flow check fuel nozzles (Ref. 73-10-05). If CT blades are eroded, refer to Inspection of CT Blades.

(a) If the nozzles are serviceable, the combustion chamber liner is the probable cause of the deposits. Inspect the CT blades, as erosion by carbon particles (some carbon particles remain in the gas stream and are not deposited in the combustion area) may damage and cause CT blade replacement.

(b) If CT blades are not eroded, engine may remain in service and the CT blades and combustion chamber liners inspected within 100 hours. Subsequent inspections are at operators discretion, but must not exceed 400 hours.

(7) As the structural integrity of the small exit duct is not affected, cracks and open radial cracks extending from the inner to the outer diameter are acceptable (length not limited), and an unlimited amount of coating loss is acceptable.

(a) Holes less than 0.500 in. in diameter in the outer wall are acceptable.

(b) If holes or open cracks are found, an HSI is recommended.

NOTE: The engine may remain in service, and HSI delayed, providing it is understood that the CT stator cooling air flow is affected, increasing the rate of deterioration.

(c) When damage is found, inspect the CT stator (Ref. 72-50-01, Inspection/Check) and associated fuel nozzle (Ref. 73-10-05, Inspection/Check).

(d) Inspect damage at 100 hours. Subsequent inspections are at the discretion of the operator, but must not exceed 400 hours. If deterioration exceeds the above limits, an HSI is recommended.

(8) Cracks along, or across, the dome to outer liner seam weld are acceptable provided the cracks do not intersect the fuel manifold support bracket. Inspect damage at 100 hours. Subsequent inspections are at the operators discretion but must not exceed 400 hours.

D. Inspection of Compressor Turbine (CT) Stator Assembly

NOTE: If borescope equipment permits keep electronic records/pictures and dimensions of observed cracks with date of inspection.

(1) Inspect the compressor turbine (CT) stator assembly per Table 603.

Table 603 Inspection of Compressor Turbine (CT) Stator Assembly

Area				Inspection
	1.	Inner Ring	(a)	Categorize multiple hairline converging cracks or parallel cracks where a piece could become detached in accordance with Figs. 610 and 611. Following criteria applies to cracks identify on inner ring or cracks extending from inner ring to airfoil surfaces:
				Inspect cracks observed per Fig. 610 as follows:

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Area				Inspection
				• Record cracks condition for size and location in engine logbook (Ref. Note) and inspect for progression at next scheduled borescope inspection. • If no progression observed, resume schedule borescope inspection interval/task. • If converging cracks extending from inner ring edge observed with progression per Fig. 610 item 1, repeat the inspection in 100 hours. Record progression observed for size and location (Ref. Note). • If converging cracks (not extending from inner ring edge) observed with progression per Fig. 610 item 2 or 3, repeat the inspection in 100 hours. Record progression observed for size and location (Ref. Note). • If cracks observed per Fig. 610 item 2 or 3 with a converged loop area smaller than 0.125 in. x 0.125 in. (or 0.0156 sq.in.), repeat the inspection in 100 hours. Record progression observed for size and location (Ref. Note).
				Inspect cracks observed per Fig. 611 as follows: • If cracks observed per Fig. 611 items with converged loop exceeding 0.125 x 0.125 in. (0.0156 sq.in.), replace/repair the CT vane within the next 100 flying hours. • If a piece of vane found missing and area exceeding 0.125 in. x 0.125 in. (0.0156 sq.in.), replace CT Vane within the next 100 flying hours and inspect compressor turbine blade for possible impact damage per Para. E.
			(b)	Multiple hairline cracks through wall of inner ring are acceptable provided 20% of axial length of inner ring remains intact:
				Record cracks condition for size and location in engine logbook (Ref. Note) and inspect for progression at next scheduled borescope inspection: • If no progression observed, resume schedule borescope inspection interval/task. • If cracks observed with progression per Fig. 610 item 4, repeat the inspection in 100 hours. Record progression observed for size and location (Ref. Note).

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Area				Inspection
				Cracks observed per Fig. 611: • If cracks are found above limits, observed per Fig. 611 item 4, replace/repair the CT vane within the next 200 flying hours.
			(c)	Open cracks up to 0.020 in. wide and 0.500 in. long are acceptable.
				Record cracks condition for size and location in engine logbook (Ref. Note) and inspect for progression at next scheduled borescope inspection: • If no progression observed, resume schedule borescope inspection interval/task. • If cracks observed with progression per Fig. 610 item 5, repeat the inspection in 100 hours. Record progression observed for size and location (Ref. Note).
				Cracks observed per Fig. 611: • If cracks are found above limits, observed per Fig. 611 item 5, replace/repair the CT vane within the next 200 flying hours.
	2.	Outer Ring	(a)	Categorize multiple hairline converging cracks or parallel cracks where a piece could become detached in accordance with Figs. 610 and 611. Following criteria applies to cracks identify on outer ring or cracks extending from outer ring to airfoil surfaces:
				Inspect cracks observed per Figure 610 as follow: • Record cracks condition for size and location in engine logbook (Ref. Note) and inspect for progression at next scheduled borescope inspection. • If no progression observed, resume schedule borescope inspection interval/task. • If converging cracks extending from outer ring edge observed with progression per Fig. 610 item 1, repeat the inspection in 100 hours. Record progression observed for size and location (Ref. Note). • If converging cracks (not extending from outer ring edge) observed with progression per Fig. 610 item 2 or 3, repeat the inspection in 100 hours. Record progression

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Area				Inspection
				observed for size and location (Ref. Note). • If cracks observed per Fig. 610 item 2 or 3 with a converged loop area smaller than 0.125 in. X 0.125 in. (or 0.0156 sq.in.), repeat the inspection in 100 hours. Record progression observed for size and location (Ref. Note).
				Inspect cracks observed per Figure 611 as follow: • If cracks observed per Fig. 611 items with converged loop exceeding 0.125 x 0.125 in. (0.0156 sq.in.), replace/repair the CT vane within the next 100 flying hours. • If a piece of vane found missing and area exceeding 0.125 in. x 0.125 in. (0.010 sq.in.), replace the CT Vane within the next 100 flying hours and inspect compressor turbine blade for possible impact damage per Para. E.
			(b)	Multiple hairline cracks through wall of outer ring are acceptable provided 20% of axial length of outer ring remains intact:
				Record cracks condition for size and location in engine logbook (Ref. Note) and inspect for progression at next scheduled borescope inspection: • If no progression observed, resume schedule borescope inspection interval/task. • If cracks observed with progression per Fig. 610 item 4, repeat the inspection in 100 hours. Record progression observed for size and location (Ref. Note).
				Cracks observed per Fig. 611: • If cracks are found above limits observed per Fig. 611 item 4, replace/repair the CT vane within the next 200 flying hours.
			(c)	Open cracks extending from outer ring edge are acceptable provided they are less than 0.020 in. wide and do not extend beyond the leading edge area.
				Record cracks condition for size and location in engine logbook (Ref. Note) and inspect for progression at next scheduled borescope inspection:

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Area				Inspection
				• If no progression observed, resume schedule borescope inspection interval/task. • If cracks observed with progression per Fig. 610 item 5, repeat the inspection in 100 hours. Record progression observed for size and location (Ref. Note).
				Cracks observed per Fig. 611: • If cracks are found above limits observed per Fig. 611 item 5, replace/repair the CT vane within the next 200 flying hours.
			(d)	Outer ring crack downstream of trailing edge area are not to be open more than 0.005 in. and not to be more than 0.500 in. long:
				Record cracks condition for size and location in engine logbook (Ref. Note) and inspect for progression at next scheduled borescope inspection: • If no progression observed, resume schedule borescope inspection interval/task. • If cracks observed with progression per Fig. 610 item 6, repeat the inspection in 100 hours. Record progression observed for size and location (Ref. Note).
				Cracks observed per Fig. 611: • If cracks are found above limits observed per Fig. 611 item 6, replace/repair the CT vane within the next 200 flying hours.
	3.	Airfoil Pressure and Suction Side	(a)	Categorize multiple hairline converging cracks or parallel cracks where a piece could become detached in accordance with Figs. 612 and 613. Following criteria applies to cracks identify on airfoil surfaces or cracks extending from airfoil surfaces to inner or outer ring.
				Inspect cracks observed per Fig. 612 as follow: • Record cracks condition for size and location in engine logbook (Ref. Note) and inspect for progression in 100 flying hours. • If no progression observed, resume schedule borescope inspection interval/task. • If converging cracks observed with progression per Fig. 612 item 1 or 2, repeat the inspection in 100 hours. Record

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Area				Inspection
				progression observed for size and location (Ref. Note).
				Inspect cracks observed per Figure 613 as follow: • If cracks observed per Fig. 613 item 1 or 2, replace/repair the CT vane at next scheduled borescope inspection. • If a piece of vane found missing and area exceeding 0.125 in. x 0.125 in. (0.0156 sq.in.), replace the CT vane at next scheduled borescope inspection and inspect compressor turbine blade for possible impact damage per Para. E.
			(b)	An open crack greater than 0.010 in. wide is not acceptable.
				Record cracks condition for size and location in engine logbook (Ref. Note) and inspect for progression at next scheduled borescope inspection. • If no progression observed, resume schedule borescope inspection interval/task. • If cracks observed with progression per Fig. 612 item 3, repeat the inspection in 100 hours. Record progression observed for size and location (Ref. Note).
				Cracks observed per Fig. 613: • If cracks are found above limits, observed per Fig. 613 item 3, replace/repair the CT vane within the next 200 flying hours.
	4.	Airfoil Leading Edge	(a)	Multiple leading edge hairline cracks extending from leading edge into each side of the airfoil up to 0.500 in. long are acceptable (no burning present).
				Record cracks condition for size and location in engine logbook (Ref. Note) and inspect for progression at next scheduled borescope inspection: • If no progression observed, resume schedule borescope inspection interval/task. • If cracks observed with progression per Fig. 612 item 4, repeat the inspection in 100 hours. Record progression observed for size and location (Ref. Note).

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Area				Inspection
				Cracks observed per Fig. 613: • If cracks are found above limits observed per Fig. 613 item 4, replace/repair the CT vane within the next 200 flying hours.
			(b)	Multiple hairline cracks (including parallel cracks) that connect to form a piece are not acceptable.
	5.	Airfoil Trailing Edge	(a)	Multiple trailing edge hairline cracks up to 1.0 inch long are acceptable (no burning present).
				Record cracks condition for size and location in engine logbook (Ref. Note) and inspect for progression at next scheduled borescope inspection: • If no progression observed, resume schedule borescope inspection interval/task. • If cracks observed with progression per Fig. 612 item 5, repeat the inspection in 100 hours. Record progression observed for size and location (Ref. Note).
				Cracks observed per Fig. 613: • If cracks are found above limits observed per Fig. 613 item 5, replace/repair the CT vane within the next 200 flying hours.
			(b)	Multiple hairline cracks (including parallel cracks) that connect to form a piece are not acceptable.
			(c)	Burn through on the vane trailing edge smaller than 0.250 in. X 0.350 in. is acceptable (Ref. Fig. 612, item 6). Repeat inspection in 100 hours. Record progression observed for size and location (Ref. Note).
			(d)	Burn through on the vane trailing edge larger than 0.250 in. X 0.350 in. is not acceptable (Ref. Fig. 613, item 6). HSI is recommended and discard the CT blades. Burned areas on the vanes increase flow area which accelerates downstream component deterioration, decreases Ng and increases T5.

(2) Inspect the CT vane for erosion and coating loss (Ref. Table 604).

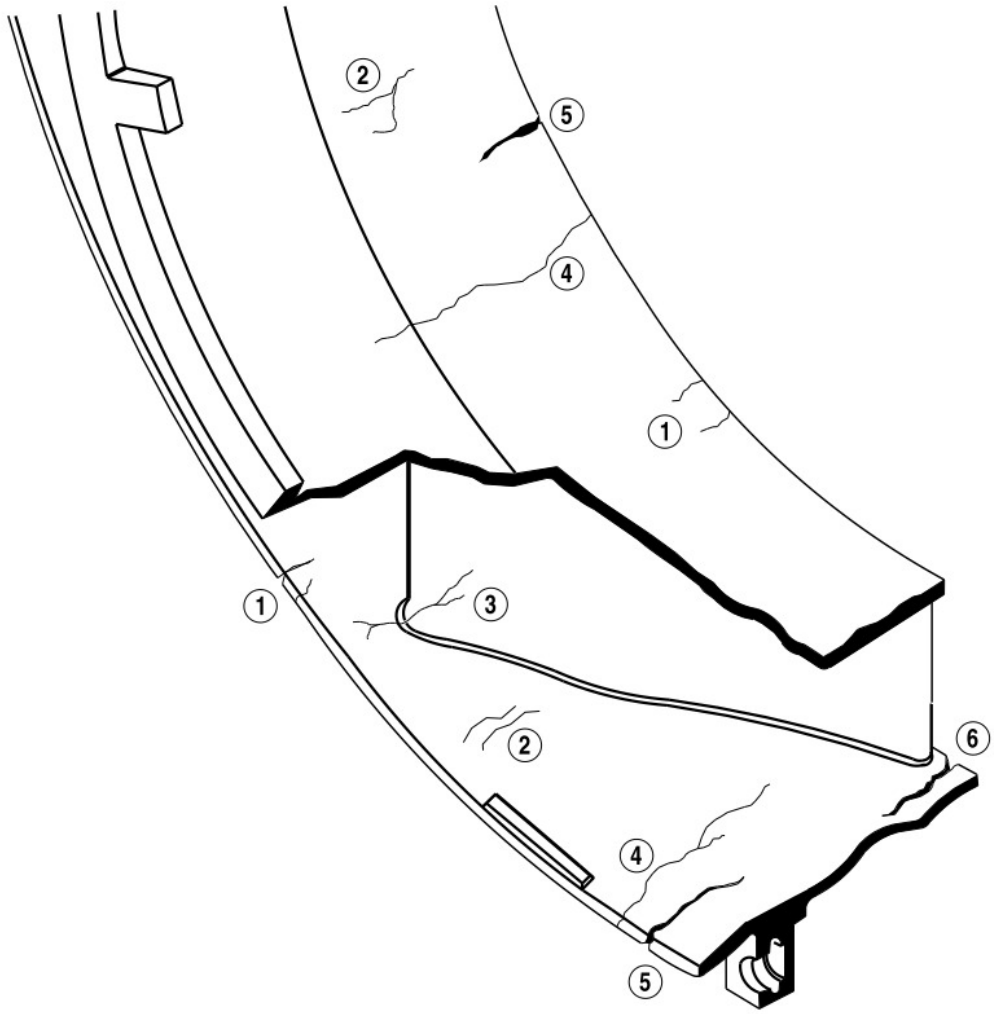
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Table 604 Inspection Criteria for Erosion and Coating Loss

Area	**Coating loss**	**Erosion**
Inner ring	Complete loss of coating is acceptable.	Any amount is acceptable.
Outer ring	Complete loss of coating is acceptable.	Any amount is acceptable.
Airfoil pressure and suction side	Any amount provided parent metal erosion is within limits.	Not Applicable

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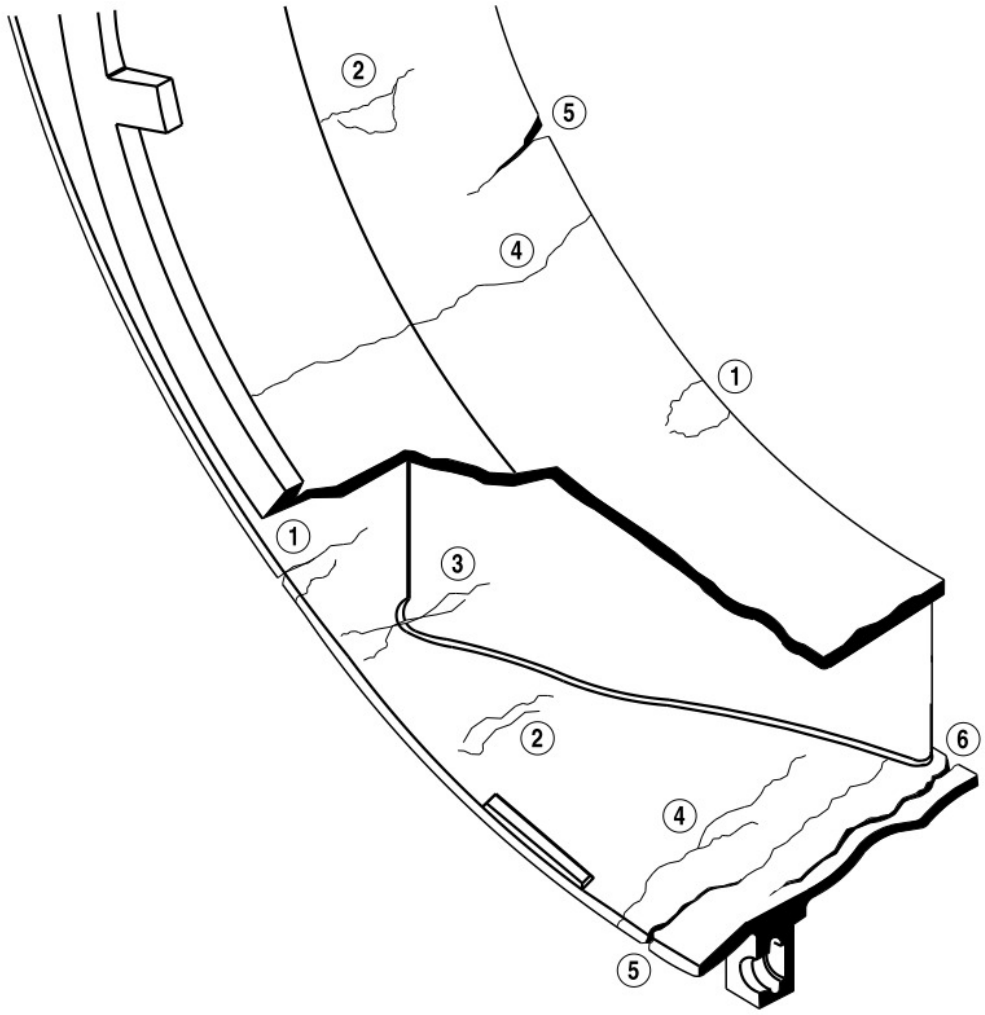
Figure 610 Acceptable Cracks on the Inner and Outer Ring



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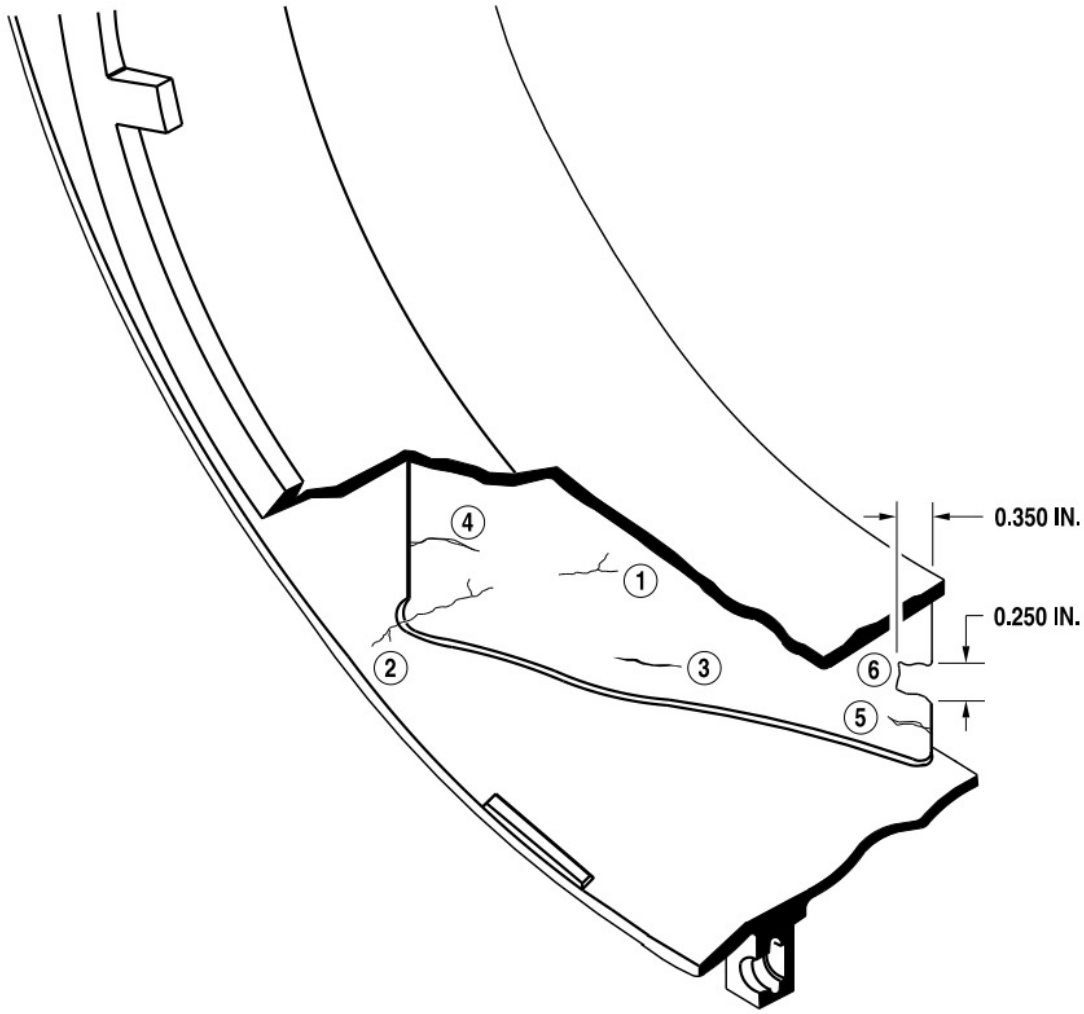
Figure 611 Not Acceptable Cracks on the Inner and Outer Ring



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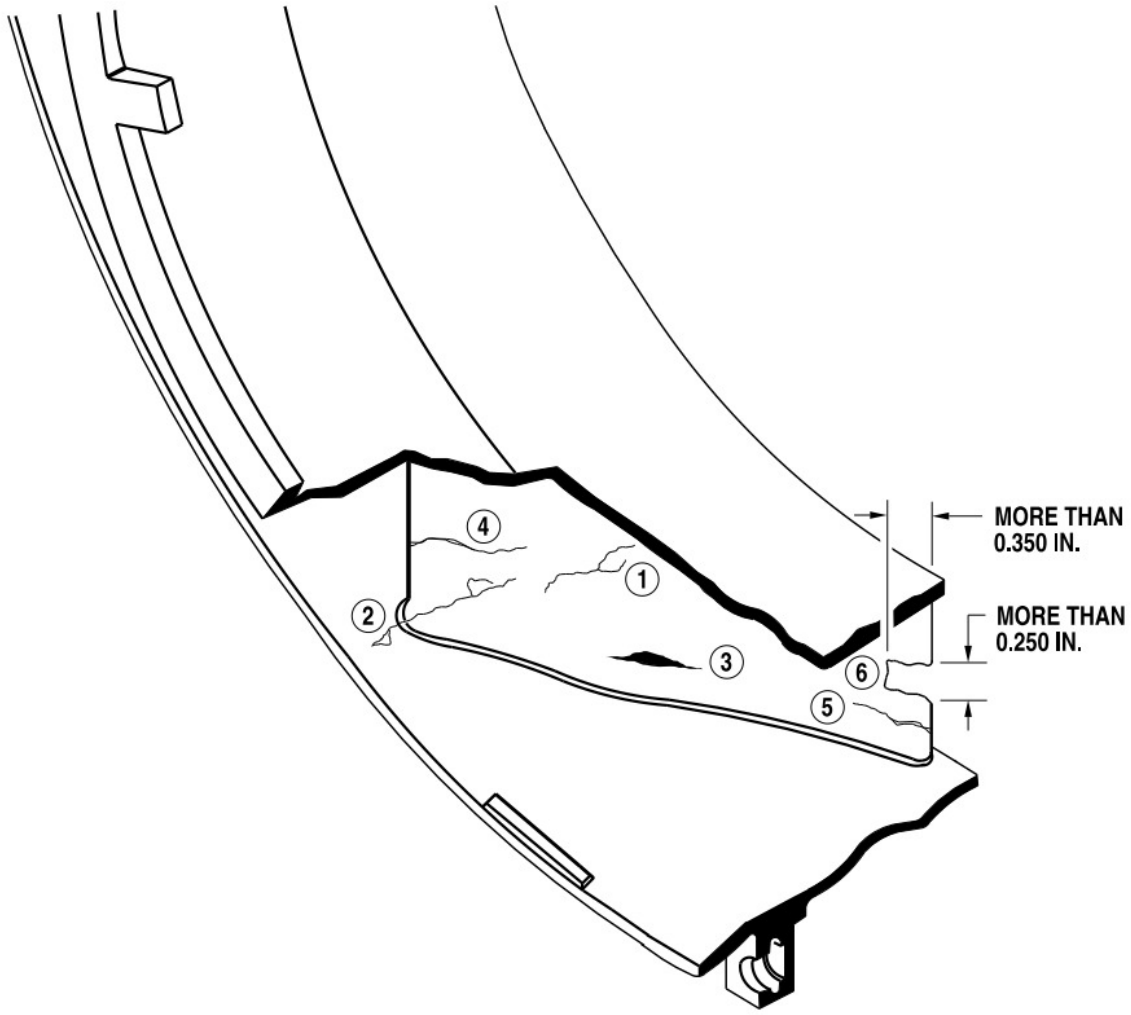
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Figure 612 Acceptable Cracks on the Airfoil



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Figure 613 Not Acceptable Cracks on the Airfoil

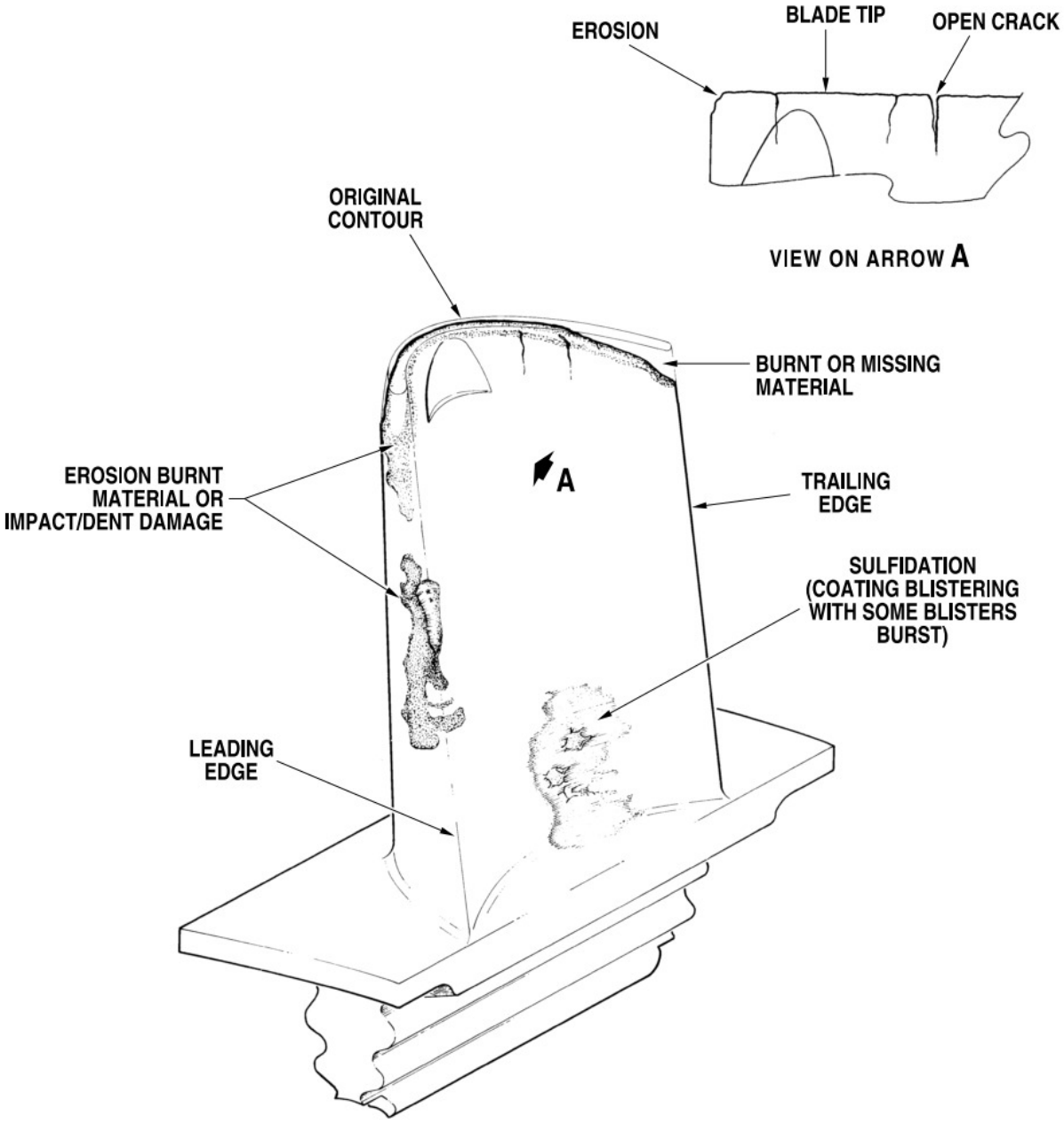


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E. Inspection of CT Blades (Ref. Figs. 614, 615 and 616)

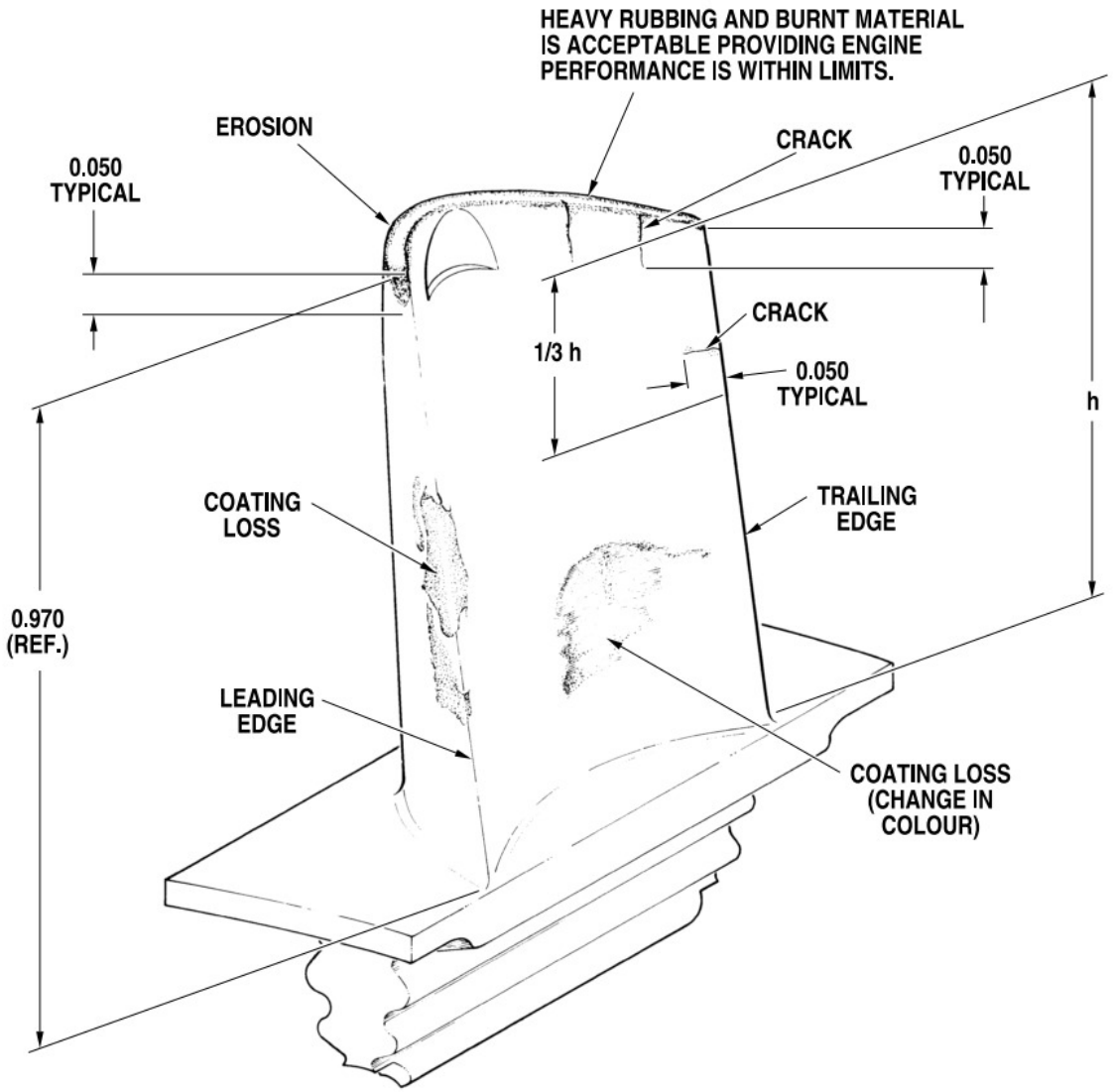
Figure 614 CT Blade Damage - HSI Required



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Figure 615 CT Blade Damage - Acceptable Damage



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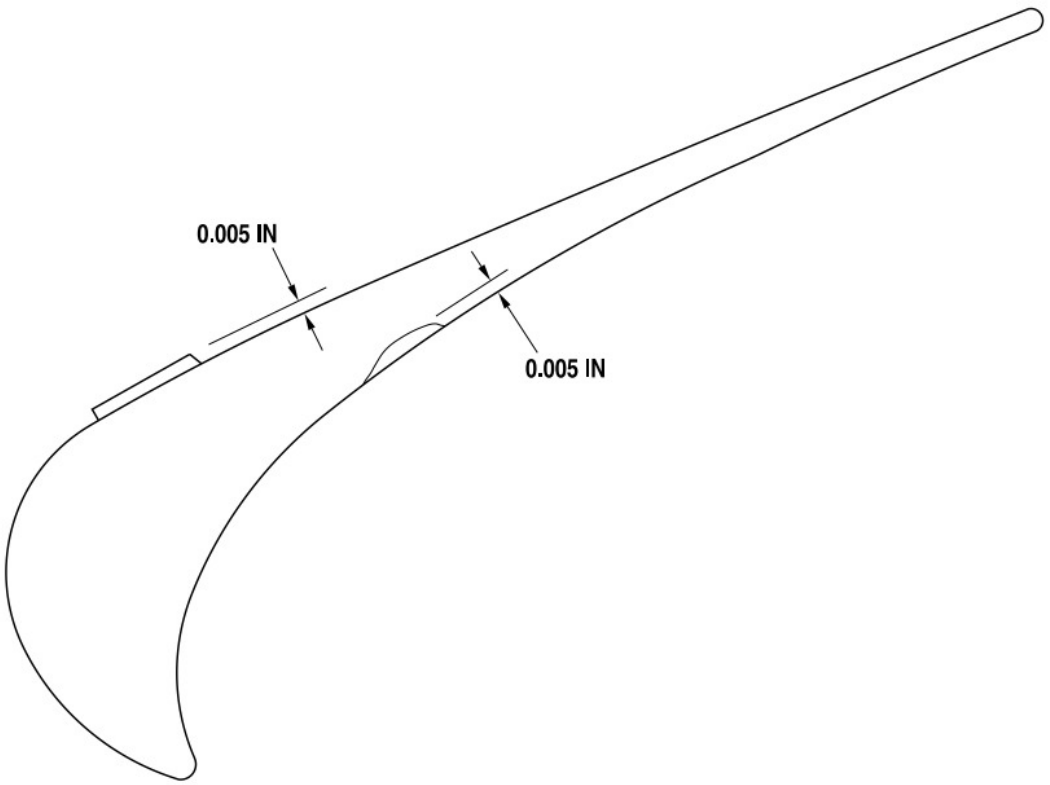
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(1) During compression turbine blade inspection, operators can find surface damages that can incorrectly identify as sulfidation. These surface damages are result of the blade manufacturing process and it will not effect turbine performance. The inspection that follow will help operators to find if the blades are serviceable:

- No loss of coating.
- No change in color.
- No signs of sulfidation on adjacent blades.
- Geometrical deviations not more than 0.005 in. deep or 0.005 in. high. There is no width limitation (Ref. Fig. 616).

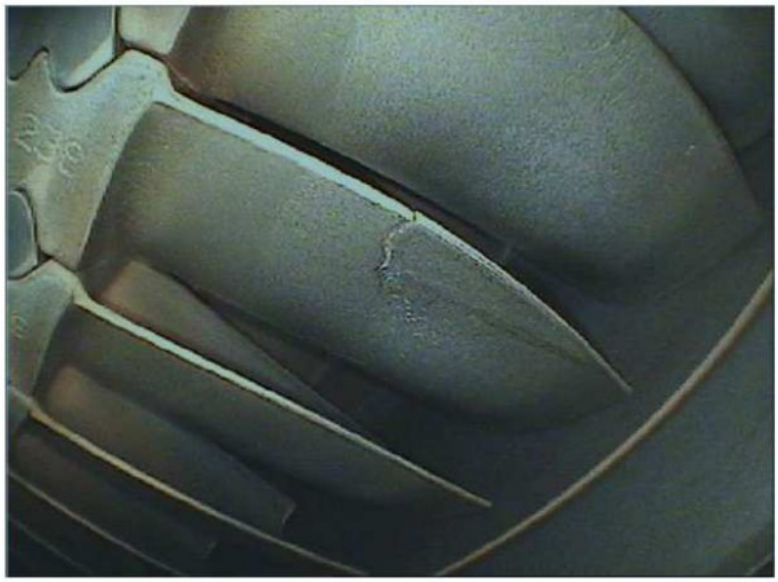
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Figure 616 CT Blade - Manufacturing Anomalies



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Figure 617 CT Blade - Trailing Edge Crack



BORESCOPE VIEW OF CT BLADES OBTAINED THROUGH THE EXHAUST DUCT PORT

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(2) The condition of blades and tips is critical to obtain rated power. Most significant blade tip damage (rubs and oxidation) increase interturbine temperature (T5). Even if T5 is below maximum, an HSI may be recommended if damage is beyond the limits shown (Ref. Fig. 615).

NOTE: Refer to Subpara. A. for additional HSI recommendations.

(a) Damages shown are acceptable, provided their condition is monitored by further inspections and engine performance checks.

(b) Subsequent inspections are at the discretion of the operator, but must not exceed 400 hours.

(c) Blades with cracks 0.050 in. long in the upper 1/3 of the trailing edge may remain in service for 100 hours.

NOTE: If nicks, dents or pits are visible in CT blades, refer to 72-50-02, COMPRESSOR TURBINE - MAINTENANCE PRACTICES.

(d) If damages are beyond those shown (Ref. Fig. 614), an HSI is recommended.

(3) Check blades for axial shift. Normally, with components within assembly tolerances, the blade platforms are approximately in line. When a blade shifts, the blade moves axially and can be seen as having moved in relation to the adjacent platform.

NOTE: 1. Checking each CT blade TE platform alignment with a borescope is quite difficult. The recommended method is to look at the leading edge (LE) mismatch while rotating the CT disk. When a mismatch is observed (one of the airfoil LE is shifted), check blade trailing edge platform. Maximum shift is 0.020 inch.

NOTE: 2. The amount of blade shift may be estimated by using a 0.020 in. thick wire wrapped tightly around the tip of borescope and held by tape. The free end should be positioned within the field of view of the borescope. When estimating the amount of shift, position the free end of the wire adjacent to the platform being checked.

(4) If blade shift in excess of the limit is observed, an HSI is recommended.

F. Inspection of CT Shroud Segments

(1) Heavy rubbing and oxidation are acceptable, providing T5 is within limits. Operating the engine with the shroud segments burned may cause damage to the CT shroud housing (cracking and burning of the attachment rim), and an HSI is recommended. A damaged shroud housing must be replaced at the next HSI or refurbishment.

NOTE: Refer to Subpara. A. for additional HSI recommendation.

G. Inspection of Power Turbine (PT) Stator

NOTE: Inspection of the PT stator is recommended when upstream component damage does not explain performance loss or when secondary damage is suspected.

(1) Damage on the vanes may produce an increase in flow area which will increase Ng and T5.

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(2) Cracks on the inner and outer rings and vanes are repairable. Keeping the engine in service, the defects will progress until the stator becomes unrepairable. This will increase the cost of the subsequent HSI. Inspect the damaged area within 100 hours. Subsequent inspections must not exceed 400 hours.

(3) An HSI is recommended when the defects are still repairable.

(4) If the defects are not repairable, replace PT stator at the next power section repair. Providing Ng and T5 are within limits, there is no need for power section repair and PT stator change, regardless of the amount of damage, unless structural integrity of the vanes are affected (e.g. wide open cracks, excessive foreign object damage (FOD) and missing or burnt material are unacceptable).

H. Inspection of PT Blades

(1) If the tip clearance of PT blades is more it can cause increase in T5. If T5 is in the limits, it is not necessary to replace the PT assembly. If there is no effect on the structural integrity of the blades, you can ignore the quantity of damage.

(a) Cracks, missing material, too much FOD, heavy sulfidation or blade distortion not permitted.

I. Inspection of Accessory Gearbox (AGB) and Reduction Gearbox (RGB) Gears

(1) The following inspection criteria should be used when inspecting the accessory and the reduction gearboxes with a borescope.

(2) This inspection may be done at operator's discretion if gearbox distress is suspected or as a troubleshooting aid as required.

(3) Acceptable Damage on Tooth Contact Faces

(a) Small clusters of spalling at one extremity of the tooth width or in a narrow band on less than 1/2 tooth length.

(b) Light shallow scoring where the scratches cannot be detected with a 0.040 in. radius scriber. Scoring usually indicates poor lubrication and the associated oil nozzle is recommended to be overhauled.

(c) Heat discoloration usually indicates poor lubrication which may lead to scoring, then spalling. The associated oil nozzle is recommended to be overhauled

(d) Ideal tooth contact is a centrally-located strip on face of tooth. Any deviation indicates incorrect tooth contact.

(e) Wear pattern starting at the end of the tooth denotes end loading. A minor amount of end loading is not considered detrimental.

(f) Incorrect tooth contact may result in spalling because the torque transmitted by the gear is transmitted by a smaller tooth surface resulting in increased contact pressure and local overloading.

(g) Components having acceptable damage may remain in service providing an inspection of the affected area is done at an interval not exceeding 400 hours. Subsequent inspections should be done at intervals depending upon the rate of progression and

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level of deterioration seen. In addition, special attention must be paid to the scheduled oil filter and chip detector inspections required by the regular maintenance program.

(4) Non-acceptable Damage

NOTE: Replace affected module or component if any of the following non-acceptable conditions are evident.

(a) Spalling in a narrow band along whole width of tooth or dispersed throughout contact area.

(b) Deep scoring (root/tip direction) across tooth contact surface.

NOTE: Deep scoring is defined as scoring where the scratches can be felt with 0.040 in. radius scriber.

(c) Rough tooth contact surface due to excessive wear, with raised material at tooth extremities.

12. Inspection of Hot Section Components

A. General

For hot section inspection frequency, refer to SB1803. For description of tasks, refer to Para. D.

Ground power check (Ref. 71-00-00, ADJUSTMENTS/TEST, Ground Power Check) before and after inspection is recommended and is important in determining the status of hot section parts and ascertain performance recovery.

For inspection of the hot section parts, refer to SB1803 (PT6A-21/27/28) and to Para. D.

The following information (Ref. Para. B. and C.) is offered as a guide to improved engine performance/handling.

B. Gas Generator Module

(1) To monitor the condition of the hot section components, a borescope may be used to inspect them (Ref. Para. 10.).

(2) Maintain CT blade radial tip clearance within prescribed limits (Ref. 72-50-01).

(3) Improve compressor turbine blade tip clearance by maintaining lug and slot joint geometry (Ref. 72-50-01, Removal/Installation) and minimizing side clearance of lugs.

(4) Maintain same CT and PT vane ring classes as installed in engine initial build or at last overhaul; that is, classes determined at engine test.

(5) Be sure the following conditions apply:

(a) Interstage sealing rings must be in satisfactory condition (Ref. 72-50-03, Inspection/Check).

(b) Compressor bleed valve must be functioning correctly.

(c) No air leaks are permitted in gas generator assembly area (e.g., cabin bleed air adapter, fuel nozzle adapter, igniter plug, P3 air lines, filter cover and adjoining lines).

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(d) Heatshield (Pre-SB1305) must be a snug fit in shroud housing.

(e) Cracks in the compressor turbine vane ring are acceptable provided they:

1 Do not exceed acceptable limits (Ref. 72-50-01).

2 Do not destroy lug and slot joint free movement and flat surface sealing.

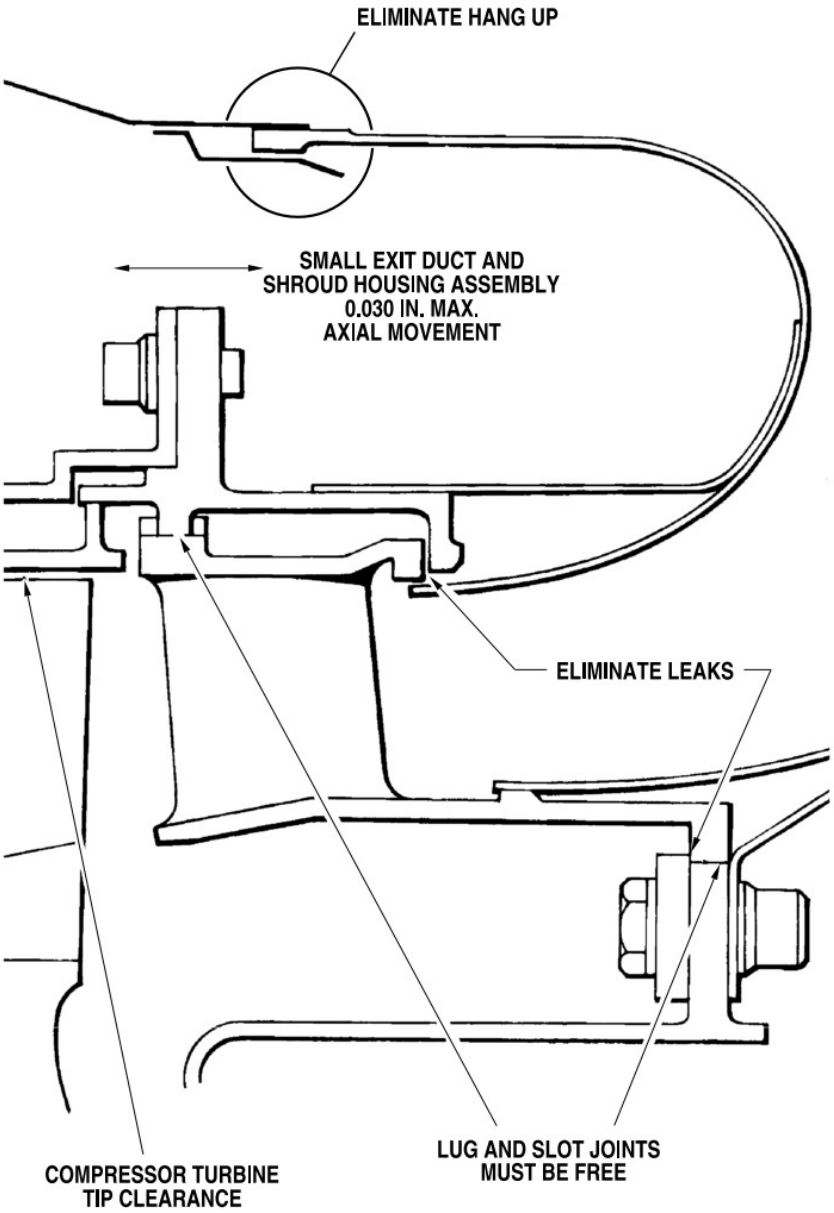
(6) The following requirements should be maintained by rework on affected components. Rework should be accomplished by hand stoning and lapping on locally manufactured lapping plates (Ref. Chapter 72-50-01, Approved Repairs):

(a) A sliding fit must exist between the combustion chamber liner and small exit duct (Ref. Fig. 618). Rework by hand.

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Figure 618 Hot Section Inspection

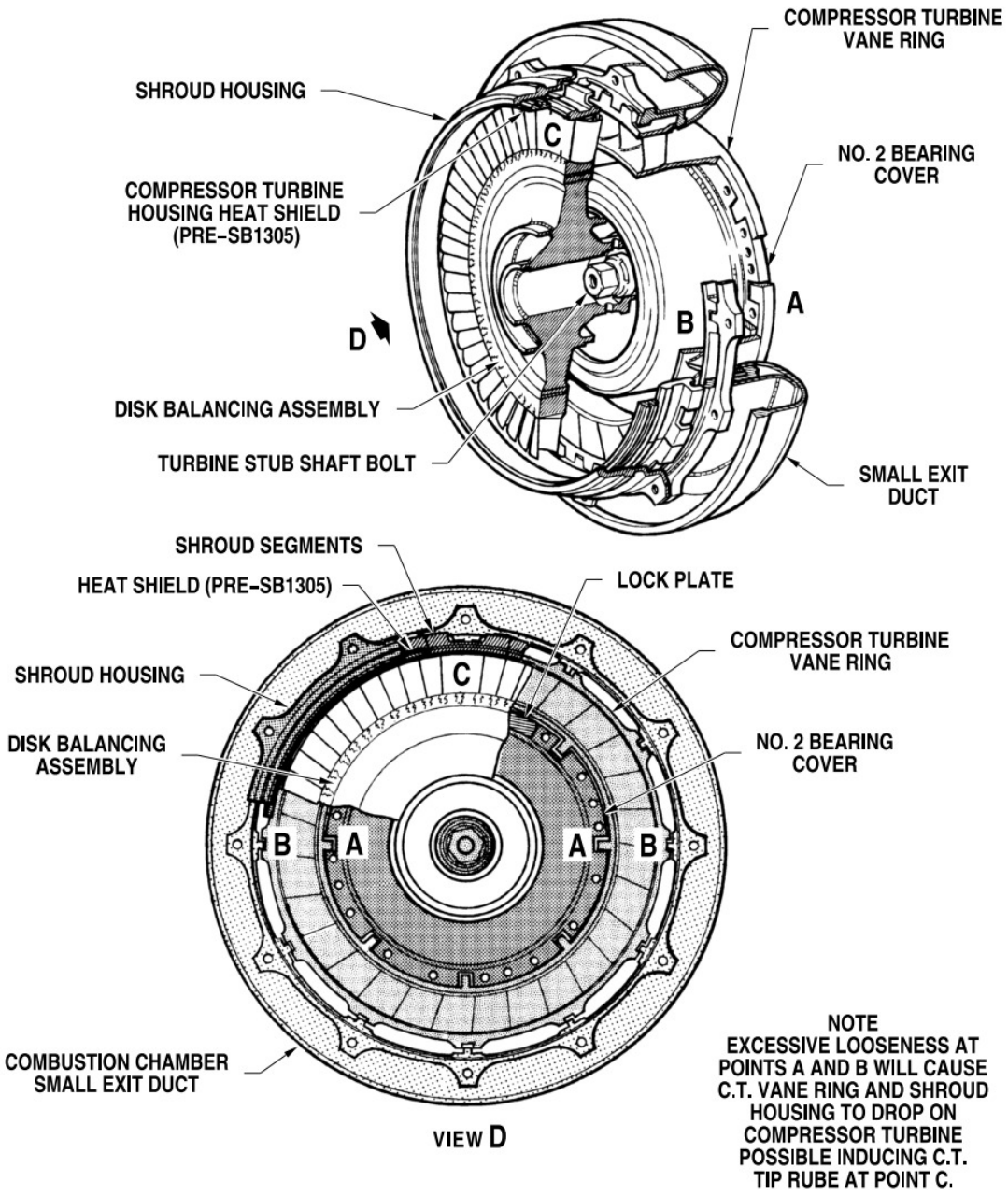
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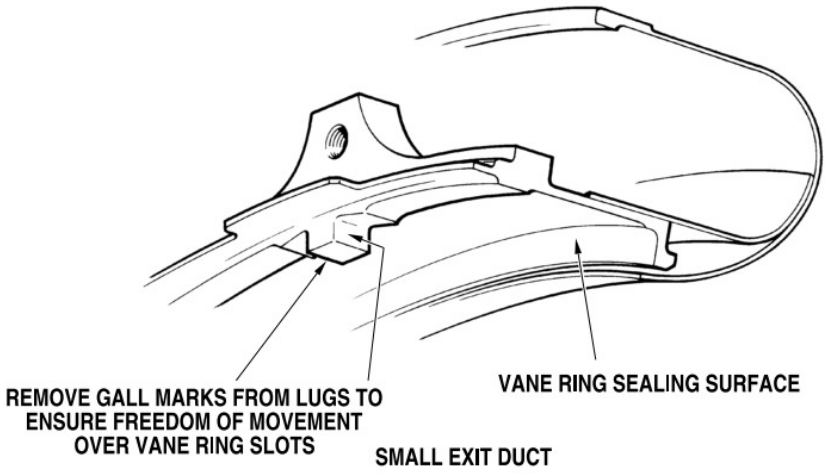
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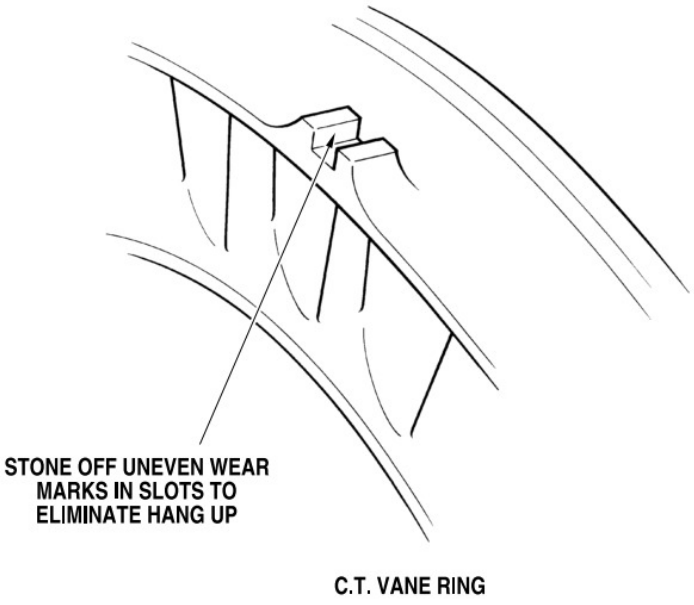


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(b) Flatness and sealing must exist on flat sealing land between small exit duct and leading edge of compressor turbine vane ring outer ring. Lap using lapping plates to obtain good sliding fit.

(c) Flatness and sealing must exist on flat sealing land between compressor turbine vane ring and lock plate. Lap as required.

(d) A free sliding fit must exist between lugs of compressor turbine vane ring and grooves in No. 2 bearing cover. Stone lugs if necessary.

NOTE: Excessive clearance between lugs and slots will cause play and destroy blade tip clearances. Replace components or return to overhaul facility for rework.

(e) Make sure compressor stator does not hang-up on CT shroud housing. Stone lugs and/or slots as necessary (Ref. Fig. 618).

C. Power Section

(1) Remove power section (Ref. REMOVAL/INSTALLATION).

(2) Remove compressor turbine disk and blade assembly (Ref. 72-50-02, Removal/Installation).

(3) Remove fuel manifold adapter assembly (Ref. 73-10-05, Removal/Installation).

(4) Remove ignition spark igniters/glow plugs (Ref. 74-20-04 or 74-20-00, Removal/Installation).

(5) Remove combustion chamber liner (Ref. 72-40-01, Removal/Installation).

NOTE: During HSI, some components listed in Table 605, remain installed. If further inspection is necessary, remove components as detailed in 72-50-01.

Table 605 Hot Section Inspection

Component	Nature of Inspection
Gas Generator Case	Cracks, distortion, overheating and corrosion (Ref. 72-30-04).
Combustion Chamber Liners, Small and Large Exit Ducts	Cracks, distortion, burning, blockage of cooling holes due to repair and coating loss. Any amount of coating loss is acceptable provided burning of parent metal has not occurred (Ref. 72-40-01, COMBUSTION CHAMBER ASSEMBLY - MAINTENANCE PRACTICES and 72-50-01, COMPRESSOR TURBINE STATOR - MAINTENANCE PRACTICES). Verify the cooling ring gaps.
Compressor Vane Ring Assembly, Shroud Housing	Cracks, coating loss, erosion of parent metal or impact damage. Examine vane ring cooling air inlet and outlet ports for blockage (Ref. 72-50-01).
Compressor	Cracks, distortion, erosion and metal build-up (Ref. 72-50-01).

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Component	Nature of Inspection
Turbine Shroud Segments	
Compressor Turbine Disk Assembly	Measure radial tip clearance (Ref. 72-50-02).
	Examine CT blades for tip rub, erosion, impact damage, coating loss, cracks, shift and circumferential movement. If a crack is found on any blade, ship the CT disk assembly to an approved overhaul facility. The complete set of blades must be discarded and replaced withCT blades (Ref. 72-50-02 and, for PT6A-28, Post-SB1265).
	Examine blade retaining rivets for condition.
Interstage Sealing Rings	Wear, fretting and distortion (Ref. 72-50-01).
Fuel Nozzles	Dissimilarity of carbon build-up. Do functional test (Ref. 73-10-05).
Fuel Nozzle Sheaths	Fretting wear, erosion and carbon build-up (Ref. 73-10-05).
Compressor Inlet	Remove air inlet screen, examine inlet area and struts, first-stage blades and vanes for dirt deposits, corrosion and cracks (Ref. 72-20-00).
T5 Temperature Sensing System	Attachment, wiring harness and lugs. Check operation (Ref. 77-20-01).
Power Section Module	
Power Turbine Stator Housing	Cracks, none permitted. Check for proper sealing with interstage sealing rings (Ref. 72-50-03).
Power Turbine Disk Assembly *	Blades for impact damage, erosion and cracks (Ref. 72-50-04).
Power Turbine Stator Assembly *	Cracks, erosion and impact damage (Ref. 72-50-03).

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Component	Nature of Inspection
Exhaust Duct	Cracks and distortion (Ref. 72-50-05).
Reduction Gearbox Oil Strainer	Remove and examine assembly for foreign matter (Ref. 72-10-00).
Chip Detector	Remove and inspect chip detector for metal deposits (Ref. 72-10-00).
* Examine in-situ.	

D. Hot Section Inspection (Ref. Fig. Appropriate HSI Sheets)

(1) For a normally scheduled HSI, do a ground performance run (Ref. 71-00-00 or AMM).

(2) Remove the RGB strainer, the main oil filter and the chip detector(s). Verify the debris (Ref. 79-20-02).

(3) Verify Compressor for FOD (Ref. 72-30-00).

(4) Remove power section/power section module (Ref. Para. 11. B.).

(5) Measure the radial CT blade tip clearance and remove compressor turbine disk and blade assembly (Ref. 72-50-02, Removal/Installation).

(6) Remove combustion chamber liner (Ref. 72-40-01, Removal/Installation).

NOTE: Should it be necessary during HSI, to make a more detailed inspection of components which remain installed, those components should be removed as detailed in 72-50-01, Inspection/Check.

(7) Verify components as per Table. 605.

(8) If HSI is done due to over temperature, remove PT stator housing (Ref. 72-50-03, Removal/Installation).

(9) If necessary, calibrate and verify aircraft instrumentation after engine assembly (Ref. AMM).

(10) Do a ground performance run (Ref. 71-00-00, POWER PLANT - ADJUSTMENT/TEST).

(11) If a significant T5 increase is noted after ground performance run, borescope hot section for CT blade shroud rub.

13. Unscheduled Inspection

NOTE: 1. If complete engine or accessory gearbox assembly is sent to an overhaul shop for repair, it is recommended to incorporate SB1564 at this time if not previously incorporated.

NOTE: 2. P&WC recommends the operator to contact the overhaul facility when an engine module is to be shipped for light overhaul or repair to confirm the engine testing requirements.

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NOTE: 3. The light overhaul workscope do not apply if the engine has gone through abnormal conditions, such as but not limited to, excessive "G" forces, abnormal temperatures that cause major material distress, exposure to chemical agents or thermal shock. This can possibly include events such as aircraft contact with the ground outside of the intended landing site, engine separation from the airframe, rotor burst, external fire or any other situation in which the aircraft receives damage that affects the structural strength, performance or flight characteristics and major repairs are necessary or replacement or repair of a major component or combination of components. In such cases, an assessment of the circumstances and the product condition is necessary to establish (if any) additional inspection, test or repairs that are necessary to make sure continued airworthiness.

NOTE: 4. The operator may be requested to provide additional information per SIL GEN-135 in order to determine the appropriate maintenance recommendations (workscope) and/or supplement existing Instructions for Continued Airworthiness. For this purpose an additional fee may apply.

NOTE: 5. It is the responsibility of the operator to supply sufficient information to the overhaul facility, such as propeller speed, power at event, extent of propeller damage, and any other damage/abnormal condition (contact with live wire) noticed on the engine/airframe for appropriate rectification. If no information is provided, the engine must be repaired to the worst case scenario.

A. General

(1) Unscheduled inspection is done when the engine is subjected to unusual stress or operating conditions, or exceeds operating limitations or gives unsatisfactory performance/handling.

(2) If as a result of the inspection, engine removal is required, a written report stating cause of removal in detail (e.g., overspeed, overtemperature, etc.) must be sent with the engine to an approved overhaul facility.

(3) Refer to the applicable accessories manual for disposition of external components or, if applicable, send component along with the removed engine or module to an approved overhaul facility.

(4) Several types of events call for rotating the propeller by hand as part of the inspection of the power section. On some engines, the power turbine may rub immediately after shutdown. This is not an indication of distress providing the rubbing/noise disappears after engine cool down.

B. Performance Deterioration

(1) Refer to FAULT ISOLATION.

C. Overspeed

(1) Check engine/aircraft indicating system. If satisfactory, do the following:

(a) If Ng exceeded 102.6 % or Np exceeded 110%, determine and rectify cause of overspeed. Send gas generator or power section to an approved overhaul facility for

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Light Overhaul, Engine Overspeed, indicating Ng or Np observed.

(b) If Np exceeded 100% for more than 20 seconds, but did not exceed 110%:

1 Inspect PT blades and check for rubs.

2 Remove and inspect RGB chip detector, oil strainer, and main oil filter (Ref. 72-10-00, 79-20-02). If metallic debris is found, refer to FAULT ISOLATION.

3 Turn the propeller by hand and listen for unusual noises coming from the reduction gearbox or PT bearings (Ref. Para. A.(4)). For unusual noises, slow propeller acceleration on start or abrupt deceleration on shutdown, send power section to an approved overhaul facility for Light Overhaul. Indicate the maximum speed attained, the duration, and any inspection discrepancy.

4 Run at 80 % TO power for ten minutes. Remove and inspect RGB chip detector, oil strainer, and main oil filter (Ref. 72-10-00, 79-20-02). If no metallic debris is found, return engine to service and check RGB chip detector daily for one week (25 hours minimum).

5 Inspect main oil filter after one week (25 hours minimum, 65 hours maximum) (Ref. 79-20-02). If no metallic debris is found, return to standard inspection intervals.

D. Inadvertent Cut-off and Relight During Taxi

NOTE: An inadvertent cut-off and relight happens when the operator moves the fuel condition lever from Low Idle to Cut-off and immediately back to Low Idle. This may result in a short-term sub-idle overtemperature.

(1) **For operators WITH an engine monitor and have obtained a detailed recording of the overtemperature event:** Refer to Chap. 71-00-00, ADJUSTMENT/TEST, Inadvertent Cut-off and Relight During Taxi figure.

(2) **For operators WITHOUT an engine monitor:** Return the compressor turbine blade and disk assembly to an overhaul shop facility for stretch check, fluorescent penetrant inspection and a metallurgical analysis (cut-up) of two blades.

E. Overtemperature

(1) Check engine/aircraft indicating system. If satisfactory, refer to Chapter 71-00-00, ADJUSTMENT/TEST for required action.

(2) When an overtemperature has occurred, and the maximum temperature reached and/or its duration cannot be established, or whenever an overtemperature is suspected to have occurred, send the engine to an overhaul facility for Light Overhaul. Indicate "Unknown Overtemperature".

NOTE: Two compressor turbine blades must be sent for metallurgical analysis (cut-up) to determine the extent of any damage to the engine.

(3) If engine was subject to an inadvertent cut-off and relight during taxi, refer to Para. D.

F. Overtorque

(1) Check engine/aircraft indicating system. If satisfactory, refer to Chapter 71-00-00,

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ADJUSTMENT/TEST and do the following .

(2) If the overtorque is in Area B:

(a) Remove and inspect the RGB chip detector, oil strainer and main oil filter (Ref. 72-10-00). If you found metallic debris, refer to FAULT ISOLATION.

(b) Turn the propeller by hand and listen for unusual noises coming from the reduction gearbox or PT bearings (Ref. Para. A.(4)). For unusual noises, slow propeller acceleration on start or abrupt deceleration on shutdown, send power section to an approved overhaul facility for Light Overhaul. Indicate torque observed and duration.

(c) Operate at 80 % TO power for ten minutes. Remove and inspect RGB chip detector, oil strainer, and main oil filter (Ref. 72-10-00). If you did not find metallic debris, back engine to servicing and examine RGB chip detector regularly for one week (25 hours minimum).

(d) Do an inspection of the main oil filter after 25 hours minimum, but not more than 65 hours maximum (Ref. 79-20-02, OIL FILTER, HOUSING AND CHECK VALVE - MAINTENANCE PRACTICES). If no metallic debris is found, return to standard inspection intervals.

(3) If the overtorque is in Area C, send PSM to an approved overhaul facility for Light Overhaul. Indicate torque observed and duration.

(4) Refer to Aircraft Maintenance Manual for related aircraft checks.

G. Immersion in Water

(1) Send engine to an approved overhaul facility for Light Overhaul. Indicate immersion in water.

H. Dropped Engine or Component

(1) Send engine or component to an approved overhaul facility for Light Overhaul. Indicate dropped engine or component; the type of surface the engine struck and from what height.

I. Material Ingestion (e.g., ice, stones, etc.)

(1) Check compressor first-stage blades for damage (Ref. 72-30-05, Inspection/Check).

(2) Do an engine performance (Ref. 71-00-00, ADJUSTMENT/TEST, Engine Performance Check) or ground power check (Ref. 71-00-00, ADJUSTMENT/TEST, Ground Power Check).

J. Bird Strike/Soft Material Ingestion (e.g., rags, plastic bags, etc.)

(1) Remains on compressor first stage blades confirm if bird or soft material has passed through the engine and possibly contaminated the gas path and CT vane cooling passages. If contamination occurred, remove the power section and inspect hot section components (Ref. In-service Inspection). Evidence of possible contamination of the compressor turbine vane cooling passages necessitates an approved overhaul level inspection of the vane.

(2) If the engine remains in service, do a performance recovery wash (Ref. 71-00-00, POWER PLANT - CLEANING, Internal Washing) and a performance (Ref. 71-00-00, ADJUSTMENT/TEST, Engine Performance Check) or ground power check as applicable

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(Ref. 71-00-00, ADJUSTMENT/TEST, Ground Power Check).

K. Chip Detector Circuit Completion and/or Debris in Oil Filter

(1) Refer to FAULT ISOLATION.

L. Propeller Sudden Stoppage or Strike (Ref. Fig. 619)

NOTE: 1. Propeller sudden stoppage occurs when propeller rotation stops because of contact (propeller strike) with a hard object (e.g. ground, ground service equipment, etc).

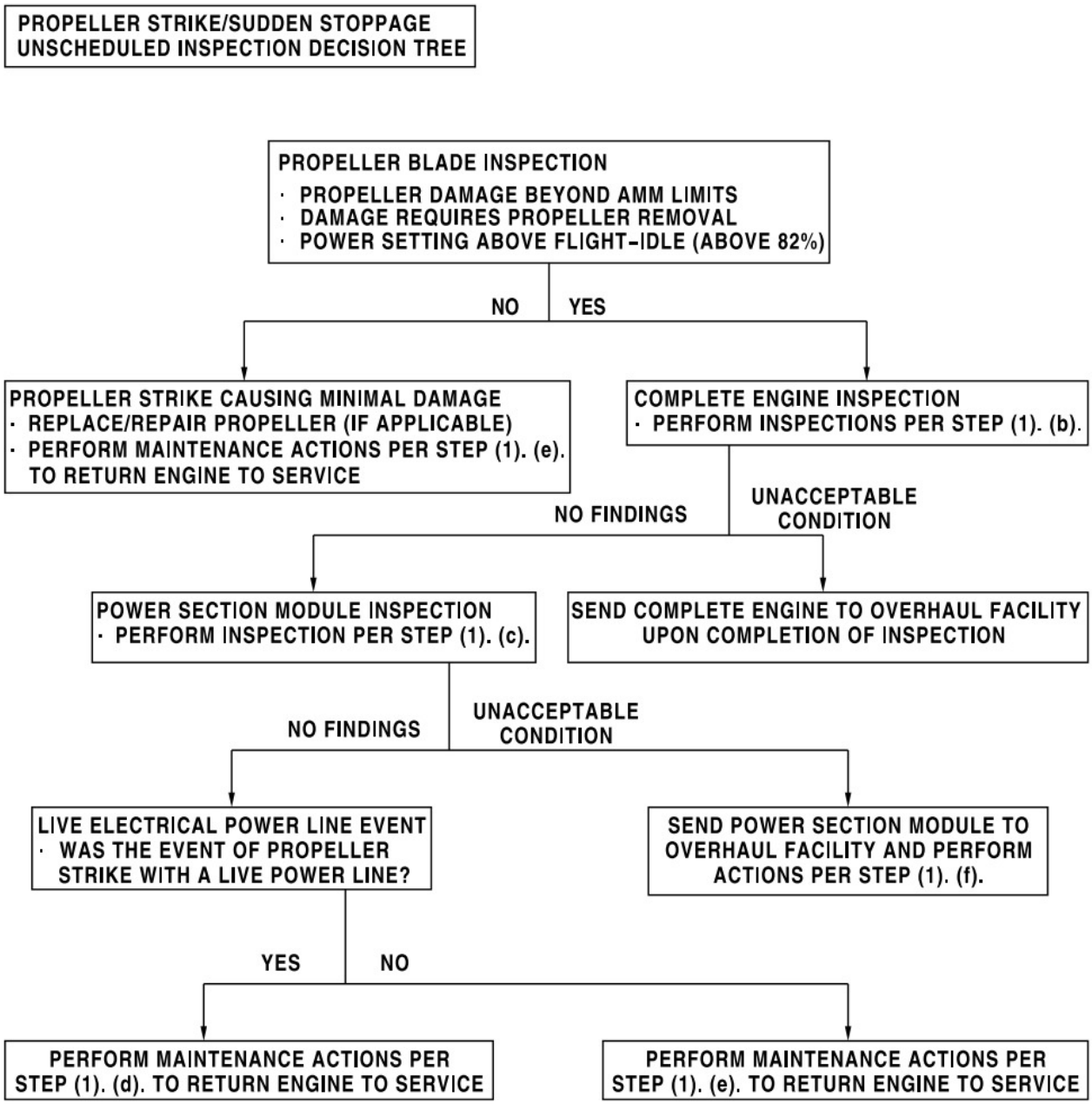
NOTE: 2. The term propeller strike is used when one of the two occurs. A rotating propeller hits an object which causes a speed variation (no stoppage) and blade damage, or a stationary propeller is hit by a moving object which causes blade damage.

NOTE: 3. P&WC recommends operators to report propeller sudden stoppage or strike events through the process outlined in P&WC Service Information Letter (SIL) GEN-135. This will enable the evaluation and investigation, as well as the determination and provision of any additional or customized maintenance recommendations that may be required to restore serviceability and reinstate warranty.

NOTE: 4. An engine removed for repair following propeller sudden stoppage or strike will be subject to the applicable light overhaul workscope in accordance with the P&WC Overhaul Manual. Due to the nature of these events and any abnormal loads or conditions, there may be exclusions for significant damage that will require specific P&WC disposition and/or additional parts replacement may be required. These conditions include but are not limited to Cracked Inlet Case Strut, Engine Separation, Change in Axial Length-to-Length, Cracked Accessories, Cracked RGB/AGB housings, Sheared Components in the Drive Train, and Substantial presence of Foreign Objects/Engine Metallic Debris.

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Figure 619 Propeller Sudden Stoppage or Strike Fault Isolation Chart



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(1) For all propeller sudden stoppage or strike, do the steps that follow:

(a) Inspect for propeller blade damage as follows:

1 For propeller damage within allowable limits (per appropriate manual) or may be repaired without propeller removal/replacement and power setting below FLIGHT IDLE (if no FLIGHT IDLE, below 82%), return engine to service following maintenance actions in Step. (1) . (e).

(b) Inspect complete engine as described in the steps below, if there are any findings of the conditions listed, remove the complete engine from service and send it to an approved overhaul facility for Light Overhaul. Indicate propeller strike and extent of damage.

1 If propeller strike was a result of a gear-up landing.

2 Casing or accessories damage, unusual noises that could indicate damage to the gears, bearings, seals or rotors, slow compressor acceleration on start or abrupt deceleration on shutdown.

3 Visually inspect for cracks or distortion at all power section module and gas generator module casings and housings, especially at or near the flanges or exhaust duct (Ref. 72-50-05, EXHAUST DUCT - MAINTENANCE PRACTICES).

4 Check all AGB and RGB installed engine and airframe components and mounting pads (Ref. 72-60-00, ACCESSORY GEARBOX ASSEMBLY - MAINTENANCE PRACTICES and 61-20-00, PROPELLER GOVERNOR - MAINTENANCE PRACTICES). Inspect for shearing, cracks, distortion or mis-alignment, and loose/pulled fasteners. Similarly, inspect all engine mounts (Ref. 72-30-04, GAS GENERATOR CASE - MAINTENANCE PRACTICES).

5 Inspect fire seals for warping or buckling (Ref. 72-30-01, CENTER FIRESEAL MOUNT RING - MAINTENANCE PRACTICE and 72-30-02, REAR FIRESEAL MOUNT RING - MAINTENANCE PRACTICES), and all external tubes for damage and/or fluid leaks. Refer to repair section or replace as necessary.

6 Check all airframe/engine connections, including fuel inlet and oil cooler lines, air system, electrical, ignition and indicating systems, control linkages, and drains (Ref. AMM).

7 For propeller strike on a power line (power utility line - does not include guide/support wires, telephone or telecommunications lines), and check for evidence of electrical discharge (localized burn marks) at the C-flange or at the compressor turbine.

8 Remove the starter-generator. Rotate the compressor by hand and listen for rubbing, scraping, interference of rotating components with stationary parts, or rapid/abrupt deceleration. Check for rear accessory case mounted accessory drag.

9 Inspect for rub on first-stage compressor rotor, housing, and stator (Ref. 72-30-05, COMPRESSOR ROTOR - MAINTENANCE PRACTICES).

10 Check for compressor turbine tip rub within limits (Ref. Para. 11.).

11 If all the above conditions are found within limits, continue with Step. (1) . (c).

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(c) Inspect power section module as described in the steps below, if there are any findings of the conditions listed, remove the power section from service and send it to an approved overhaul facility for Light Overhaul. Indicate propeller strike and extent of damage.

1 Events where power setting was above FLIGHT-IDLE (or 82% if FLIGHT IDLE not defined).

2 For propeller damage beyond this criteria:

a The blade is loose in the hub

b For a metal blade, the blade is bent out of track or angle, or the blade tip is missing more than 1 in., or

c For a composite blade, the spar is damaged or exposed, or the tip is damaged more than 5 in.

3 Turn the propeller by hand and listen for unusual noises coming from the reduction gearbox or PT bearings (Ref. Para. A. (4) .).

4 As described below unusual noises, vibration, slow propeller acceleration on start or abrupt deceleration on shutdown. remove the power section from service and send it to an approved overhaul facility for Light Overhaul. Indicate propeller strike and extent of damage.

5 Was there any vibration reported that was not present before the event.

6 Perform steps in accordance with Step. (1) . (f). for events requiring power section module replacement only.

7 If the conditions above are found within limits and propeller strike occurred with a live power line proceed to Step. (1) . (d)., otherwise proceed to Step. (1) . (e).

(d) Propeller strike causing electrical damage:

1 For propeller strike on a live power line (power utility line - does not include guide/support wires, telephone or telecommunications lines), or propeller strikes causing electric discharge (short circuit), check for evidence of electrical discharge (localized burn marks) at the propeller shaft, RGB housing, AGB housing, starter generator, accessories and exhaust duct. Refer to Propeller Lightning Strike Inspection and Propeller Electrical Leads Shorting Inspection (Ref. Para. 13. M.).

2 Continue with Step. (1) . (e).

(e) Propeller strikes causing minimal damage; or for propeller strikes with NO findings per conditions listed in inspection procedures for the propeller, engine, power section module and electrical damage check (Ref. Step. (1) . (b)., Step. (1) . (c)., or Step. (1) . (d).), and for events where power setting was below FLIGHT-IDLE (if no FLIGHT IDLE, below 82%); or engine was shut down; proceed with the following maintenance actions to return engine to service:

1 Remove and inspect RGB chip detector, oil strainer, and main oil filter (Ref. 72-10-00, REDUCTION GEARBOX - MAINTENANCE PRACTICES and Ref. 79-20-02, OIL FILTER

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HOUSING AND CHECK VALVE - MAINTENANCE PRACTICES). If metallic debris is found, refer to FAULT ISOLATION.

2 Run engine at 80% TO power for ten minutes. Remove and inspect RGB chip detector, oil strainer, and main oil filter (Ref. 72-10-00, REDUCTION GEARBOX - MAINTENANCE PRACTICES and Ref. 79-20-02, OIL FILTER HOUSING AND CHECK VALVE - MAINTENANCE PRACTICES). If no metallic debris is found, return engine to service and check RGB chip detector daily for one week (25 hours minimum).

3 Inspect main oil filter after one week (25 hours minimum, 65 hours maximum) (Ref. 79-20-02, OIL FILTER HOUSING AND CHECK VALVE - MAINTENANCE PRACTICES). If no metallic debris is found, return to standard inspection intervals.

(f) For events requiring power section module replacement only:

CAUTION: WHEN INSTALLING A REPLACEMENT POWER SECTION MODULE (PSM) ONTO A GAS GENERATOR MODULE (GGM), MAKE SURE THE APPROPRIATE POWER TURBINE STATOR ASSEMBLY IS INSTALLED IN THE PSM TO MAINTAIN ORIGINAL MATCHING AND THE PROPER T5 IMMERSION THERMOCOUPLES ARE INSTALLED TO SUIT THE CONFIGURATION OF THE GGM COMBUSTION CHAMBER LINER.

1 Inspect the power turbine stator (Ref. 72-50-04, POWER TURBINE - MAINTENANCE PRACTICES), specifically for evidence of blade tip rub or contact. If within limits, keep the power turbine stator with the installed gas generator module.

2 After installation of a repaired/replacement power section on the serviceable gas generator, run at 80% TO power for ten minutes. Remove and inspect the main oil filter (Ref. 79-20-02, OIL FILTER, HOUSING AND CHECK VALVE - MAINTENANCE PRACTICES). If no metallic debris is found, return the engine to service.

3 Inspect main oil filter after one week (25 hours minimum, 65 hours maximum) (Ref. 79-20-02, OIL FILTER HOUSING AND CHECK VALVE - MAINTENANCE PRACTICES). If no metallic debris is found, return to standard inspection intervals.

M. Propeller Lightning Strike

(1) If signs of arcing are found on the propeller blades, the propeller shaft and flange must be checked for magnetism. Check for magnetism using one of the following methods:

(a) Check propeller shaft, especially flange, using a magnetometer to make sure residual magnetism is within ± 3 gauss or 3 oersted of 2.38 A/cm. If reading is above limits, replace power section.

(2) If propeller shaft or flange is magnetized:

(a) Ship power section to an approved overhaul facility for Light Overhaul. Indicate Lightning Strike.

(b) On the gas generator, check for evidence of electrical discharge (localized burn marks) at the C-flange or at the compressor turbine. Evidence of discharge requires sending

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the complete engine to an approved overhaul facility.

(c) Inspect the main oil filter (Ref. 79-20-02). If metallic debris is found, refer to FAULT ISOLATION.

(3) If the propeller shaft or flange is not magnetized:

(a) Remove and inspect the RGB chip detector, oil strainer, and main oil filter (Ref. 72-10-00). If you found metallic debris, refer to FAULT ISOLATION.

(b) Turn the propeller by hand and listen for unusual noises from the reduction gearbox or PT bearings (Ref. Para. A.(4)). For unusual noises, slow propeller acceleration on start or abrupt deceleration on shutdown, send power section to an approved overhaul facility for Light Overhaul and indicate lightning strike.

(c) Operate engine at 80% TO power for ten minutes. Remove and inspect the RGB chip detector, oil strainer, and main oil filter (Ref. 72-10-00). If you did not find metallic debris, back the engine to servicing and examine RGB chip detector regularly for one week (25 hours minimum).

(d) Inspect main oil filter after one week (25 hours minimum, 65 hours maximum) (Ref. 79-20-02). If no metallic debris is found, return to standard inspection intervals.

N. Propeller Electrical Leads Shorting

(1) If signs of arcing between de-icer leads and the propeller spinner, bulkhead or hub assembly have been found, check propeller de-icing system (Ref. Aircraft Maintenance Manual) and rectify.

(2) Inspect engine:

(a) Remove and inspect RGB chip detector, oil strainer and main oil filter (Ref. 72-10-00). If you found metallic debris, refer to FAULT ISOLATION.

(b) Turn the propeller by hand and listen for unusual noises coming from the reduction gearbox or PT bearings (Ref. Para. A.(4)). For unusual noises, slow propeller acceleration on start or abrupt deceleration on shutdown, send power section to an approved overhaul facility for Light Overhaul. Indicate propeller electrical leads short and any discrepancy noted. Otherwise, run engine at 80% TO power for ten minutes.

(c) Remove and inspect RGB chip detector, oil strainer, and main oil filter (Ref. 72-10-00). If you did not find metallic debris, back engine to servicing and examine RGB chip detector regularly for one week (25 hours minimum).

(d) Inspect main oil filter after one week (25 hours minimum, 65 hours maximum) (Ref. 79-20-02). If no metallic debris is found, return to standard inspection intervals.

O. Heavy/Hard Landing

NOTE: 1. Do the following checks in the event of a suspected heavy or hard landing. A landing must be considered hard if the aircraft incurred any airframe or landing gear damage.

NOTE: 2. If the heavy/hard landing involved a windmilling (inoperative) engine, send the

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complete engine to an approved overhaul facility for Light Overhaul. Indicate heavy/hard landing of inoperative engine. Otherwise, proceed as follows:

(1) Inspect for cracks or distortion at all engine casings and housings, especially at or near the flanges.

(2) Remove the starter-generator and check all AGB pads and AGB mounted engine and airframe components. Look for shearing, cracks, distortion or mis-alignment, and loose/pulled fasteners. Similarly, inspect all other accessories and engine mounts.

(3) Inspect fireseals for warping or buckling, and all external tubes for damage and/or fluid leaks.

(4) Check all airframe/engine connections, including fuel inlet and oil cooler lines, air system, electrical, ignition and indicating systems, control linkages, and drains.

(5) Remove and inspect RGB chip detector, oil strainer, and main oil filter (Ref. 72-10-00). If you found metallic debris, refer to FAULT ISOLATION.

(6) Rotate the compressor by hand and listen for rubbing, scraping, interference of rotating components with stationary parts, or rapid/abrupt deceleration. Check for rear accessory case mounted accessory drag. Check for compressor turbine tip rub (Ref. 72-50-02, Inspection/Check). Rotate the propeller by hand and listen for any interference of rotating components with stationary parts or any other unusual noises coming from the gearbox or turbines (Ref. Para. A.(4)). Examine PT disk assembly and exhaust area through exhaust ports for evidence of distress. If damage is found, inspect (Ref. 72-50-04, Inspection/Check).

(7) For casing or accessories damage, unusual noises that could indicate damage to the gears, bearings, seals or rotors, slow propeller acceleration on start or abrupt deceleration of compressor or power section on shutdown, send the complete engine to an approved overhaul facility for Light Overhaul. Indicate heavy landing and discrepancies observed.

(8) Operate the engine at 80% TO power for ten minutes. Remove and inspect RGB chip detector, oil strainer, and main oil filter (Ref. 72-10-00). If you did not find metallic debris, back engine to servicing and examine RGB chip detector regularly for one week (25 hours minimum).

(9) Inspect main oil filter after one week (25 hours minimum, 65 hours maximum). If no metallic debris is found, return to standard inspection schedule.

P. Aircraft Flown Through Volcanic Ash or Smoke

(1) Wash compressor and turbine (Ref. 71-00-00, POWER PLANT - CLEANING, Internal Washing).

(2) Drain and refill oil system with new oil (Ref. SERVICING, Lubricating Oil System).

(3) Clean or change main oil filter (Ref. 79-20-02, CLEANING).

(4) Examine compressor (Ref. INSPECTION/CHECK, Borescope).

(5) Return engine to service if no defects are found.

(6) Drain and refill oil system with new oil (Ref. SERVICING, Lubricating Oil System) 50 ± 10 flight

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hours after original oil change done in step (2).

Q. Sustained Running at Oil Temperature Outside Limits

(1) Check aircraft/engine indicating system and correct cause of high oil temperature (Ref. FAULT ISOLATION).

(2) Rotate the compressor rotor and check for indications of AGB or bearing distress.

(3) Turn the propeller by hand and listen for unusual noises coming from the reduction gearbox or PT bearings (Ref. Para. A.(4)).

(4) For any presence of unusual noise, send the complete engine to an approved overhaul facility for Light Overhaul. Indicate the oil temperature that the engine was operated at, the engine power and the duration.

(5) Drain and discard oil (Ref. SERVICING, Lubricating Oil System).

(6) Remove and inspect the RGB chip detector, oil strainer, and main oil filter (Ref. 72-10-00). If you found metallic debris, refer to FAULT ISOLATION.

(7) Fill oil system (Ref. SERVICING, Lubricating Oil System).

(8) Operate the engine at 80% TO power for ten minutes. Remove and inspect RGB chip detector, oil strainer, and main oil filter (Ref. 72-10-00). If you did not find metallic debris, back engine to servicing and examine RGB chip detector regularly for one week (25 hours minimum).

(9) Inspect main oil filter after one week (25 hours minimum, 65 hours maximum) (Ref. 79-20-02). If no metallic debris is found, return to standard inspection intervals.

R. Loss of Oil/Oil Pressure or Low Oil Pressure

CAUTION: IF AIRCRAFT OIL SYSTEM CONTAMINATION IS SUSPECTED, REVERSE FLUSH ENGINE OIL SYSTEM, INCLUDING OIL-TO-FUEL HEATER. AIRCRAFT OIL COOLER MUST BE REJECTED.

NOTE: Low oil pressure is defined as running the engine with the oil pressure below limits (Ref. 71-00-00, Table 503).

(1) If the loss of oil is 6 qt. or more, resulting in fluctuation of oil pressure or torque indication or oil pressure dropped below minimum value.

(a) Check oil pressure and torque indicating system. If correct, and the engine did not run above 1315 lb.ft. (PT6A-21), 1628 lb.ft. (PT6A-27/28) torque with oil pressure below 80 psig (minimum time running is permitted to allow flight crew to follow the emergency procedure), remove main filters, strainer and chip detector. If metallic debris is found refer to 79-20-04, Inspection.

NOTE: 1. Oil pressures below 40 psig are unsafe and require either the engine to be shut down or an emergency landing made as soon as possible, using minimum power to sustain flight.

NOTE: 2. If oil pressure fluctuates or drops below 80 psig, reduce the engine power from the maximum of 1315 lb.ft. (PT6A-21), 1628 lb.ft. (PT6A-27/28) torque.

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The engine may be kept running provided that the oil pressure does not fall lower than 40 psig; however the torque indicating system will be unreliable at this low pressure. Minimum time running is permitted to allow the flight crew to follow emergency procedure.

(b) Turn the propeller by hand and listen for unusual noises coming from the reduction gearbox or PT bearings. If unusual noises are heard, send power section to an approved overhaul facility for Light Overhaul. Indicate unusual oil conditions.

NOTE: PT blades may rub, immediately after engine shutdown which is not an indication of distress, providing the rubbing/noise disappears after engine cool down.

(c) Turn the compressor rotor. Listen for unusual noises coming from bearings, seals, gears, compressor and/or CT.

(d) If unusual noises are heard, send gas generator to an approved overhaul facility for Light Overhaul. Indicate unusual oil conditions.

(e) Remove main oil filter, strainer and chip detector and inspect for metallic debris. If metallic debris is found, refer to FAULT ISOLATION. If no metallic debris is found, reinstall main oil filter, strainer and chip detector. Run the engine at 80% TO power for 10 minutes; recheck RGB chip detector, oil strainer and main oil filter. If no metallic debris is found return engine to service. Inspect chip detector daily for one week (approx. 65 flight hours) and main oil filter after seven days (approx. 15 flight hours). If no metallic debris is found, return to standard inspection intervals. If metallic debris is found, refer to FAULT ISOLATION.

(2) If the engine was kept running above 1315 lb.ft. (PT6A-21), 1628 lb.ft. (PT6A-27/28) torque with oil pressure below 80 psig (Ref. 71-00-00, Table 503), check oil pressure and torque indicating systems; and if correct, remove the engine and send it to an overhaul facility for Light Overhaul. Indicate low oil pressure.

(3) If the loss of oil is 6 qt. or more, and the engine oil pressure or torquemeter indication have fluctuated or oil pressure dropped below 40 psig:

NOTE: If the oil pressure drops below 40 psig, an engine shut down is required. In single engine installations or in emergency, use minimum power to sustain flight and prepare for an emergency landing.

(a) Check oil pressure and torque indicating system. If correct, and the engine was kept running with oil pressure below 40 psig in excess of the time required to carry out the emergency engine shutdown procedure, ship the engine to an overhaul facility for Light Overhaul. Indicate unusual oil conditions.

(b) Remove and ship propeller governor for overhaul. Indicate loss of oil.

S. Oil Pressure Follows Throttle

(1) Under normal operating conditions, oil pressure indication is stable. Malfunction of the pressure relief valve or oil leakage will cause indication to increase as throttle is advanced, or drop when throttle is retarded.

(a) Check external lines for breaks and leakage. Remove pressure relief valve and check for

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sticking, scoring, etc.

(b) Remove accessory gearbox. Check oil pump housing for cracks.

(c) Remove relief valve and housing. Examine housing for wear resulting from relief valve rotation. Replace housing if necessary.

T. Contamination by Fire Extinguishing Agents

(1) Foam, powder or other chemical extinguishers:

(a) If only engine externals are exposed, then wash (Ref. 71-00-00, CLEANING, Engine External Wash) using drinkable quality water only and monitor for corrosion.

(b) In the event of internal contamination, return engine to an approved overhaul facility for Light Overhaul. Indicate the agent ingested by the engine. Depending on the agent, and at the option of the operator, an attempt may be made at cleaning the contaminating agent prior to shipping:

1 Do a dry motoring run (Ref. 71-00-00, ADJUSTMENT/TEST, Dry Motoring Run) to blow out any residual deposits.

2 Wash engine externally (Ref. 71-00-00, CLEANING) using drinkable quality water only.

3 Do an engine performance recovery wash (Ref. 71-00-00, CLEANING) followed by two dry motoring runs.

(2) Halon:

(a) No engine maintenance required.

U. Audible Rubbing, Binding or Scraping

NOTE: Any unusual engine noise requires immediate investigation.

(1) Rotate compressor rotor and listen for any interference of rotating components with stationary parts and/or check for rapid deceleration on shutdown.

(a) Do a visual inspection of the first-stage compressor rotor with a borescope or with the air inlet screen removed (Ref. 72-30-05, COMPRESSOR ROTOR - MAINTENANCE PRACTICES).

(b) Check for rear accessory case mounted accessory drag.

(c) Check for compressor turbine tip rub (Ref. 72-50-02, Inspection/Check).

(d) Check for the indications of compressor rub or bearing or AGB distress or unusual deposits seen on CT blade airfoil. Send engine to an approved overhaul facility for light overhaul.

NOTE: If you find deposits, look for the presence of debris at the bottom of the gas generator case. If you see one of these conditions, contact P&WC Customer Engineering for recommendations.

(e) If no damage is found and condition is still present for one hour after shutdown, send the engine to an approved overhaul facility.

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(2) Rotate the propeller and listen for any interference of rotating components with stationary parts (Ref. Para. A.(4)). If power turbine rotor rattles, or if there is rubbing or scraping, slow propeller acceleration on start or abrupt deceleration on shutdown:

(a) Examine PT disk assembly and exhaust area through exhaust ports for evidence of distress.

(b) If damage is found, inspect (Ref. 72-50-04, Inspection/Check).

(c) Slow or stiff propeller rotation. Send power section to an approved overhaul facility for Light Overhaul. Indicate rubbing.

(3) If the suspected problem cannot be repeated or confirmed:

(a) Remove and inspect RGB chip detector, oil strainer, and main oil filter (Ref. 79-20-02). If metallic debris is found, refer to FAULT ISOLATION.

(b) Run engine at 80% TO power for ten minutes. Remove and inspect RGB chip detector, oil strainer, and main oil filter (Ref. 79-20-02). If no metallic debris is found, return engine to service and check RGB chip detector daily for one week (25 hours minimum).

(c) Inspect main oil filter after one week (25 hours minimum, 65 hours maximum) (Ref. 79-20-02). If no metallic debris is found, return to standard inspection intervals.

V. Propeller Windmilling after In-flight Shutdown

(1) Investigate cause of shutdown. If suspected of being attributed to an engine or accessory problem, send engine or accessory to an approved overhaul facility for Light Overhaul. Indicate engine shutdown and the events and conditions at the time.

(2) With shutdown not related to an engine problem (serviceable engine):

(a) With stabilized windmilling Np not more than 20 rpm and less than 6 qt. of oil required to bring oil level to MAX on dipstick, no further action is necessary.

(b) With an unknown (not recorded) Np or stabilized windmilling Np greater than 20 rpm or if 6 qt. or more of oil is required to bring oil level to MAX on dipstick:

1 Remove and inspect RGB chip detector, oil strainer, and main oil filter (Ref. 79-20-02). If metallic debris is found, refer to FAULT ISOLATION.

2 Rotate the compressor rotor and check for indications of AGB or bearing distress.

W. Contamination of Oil with Non-metallic Foreign Material

NOTE: 1. AGB/RGB internal protective coatings may be released within the engine and appear as flakes 1/64 inch to 3/8 inch in diameter, usually shiny, yellow, brown or green in color, and may or may not be transparent.

NOTE: 2. Inspect chip detector, main oil filter and AGB scavenge pump screen after approximately 10 hours. If no debris is found, inspect at 50 hours.

NOTE: 3. If no debris is found at 50 hours, refer to the standard periodic inspection interval for

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oil filter and chip detector (Ref. Table 601).

NOTE: 4. If these flakes are found, immediately contact your local P&WC customer support representative. Send engine to an approved overhaul facility for Light Overhaul. Indicate oil contamination from released AGB/RGB internal protective coatings.

(1) Remove chip detector from reduction gearbox front housing (Ref. SERVICING,Oil Change) and scrub scavenge oil strainer sleeve with a brush to remove any foreign material blocking the screen.

(2) Remove accessory gearbox drain plug(s).

(3) Flush oil system (Ref. SERVICING, Oil System Flushing).

(4) Reinstall chip detector and plug(s) and tighten. Refill engine oil tank.

NOTE: Do not lockwire chip detector and drain plug(s) at this time.

(5) Remove chip detector and inspect scavenge oil strainer sleeve for residual foreign material accumulated after system flush (Ref. NOTE 1).

(6) Repeat steps (2) thru (5) , if required.

(7) Reinstall chip detector. Tighten all plugs and chip detector and lockwire.

X. Inspection of Main Oil Filter Secondary Screen (Ref. Fig. 620)

Figure 620 Inspection of Main Oil Filter Secondary Screen



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(1) If you find a tear or opening on the main oil filter secondary screen (coarse hat-type screen attached to the inner end of the filter), do a main oil filter patch check (Ref. 79-20-02, OIL FILTER HOUSING - MAINTENANCE PRACTICES).

(2) If you find contamination, send the engine to an approved overhaul facility for light overhaul. Give the type of contamination, as applicable (example: carbon, metallic, non-metallic, etc).

Y. Starter-Generator Replacement

(1) If there is an engine starting fault or an electrical generation defect, do the following inspections:

(a) Inspect the starter-generator drive splines:

1 Inspect per Aircraft Maintenance Manual (AMM).

2 If there is no inspection procedure in the AMM Chapter 80-10-00, clean the starter-generator splines and inspect for evidence of electrical arcing damage (in the form of pitting) using a 10X magnifying glass.

NOTE: 1. If there is any evidence of electrical arching, return engine to an approved overhaul facility for Electrical Discharge Damage (EDD) repair.

NOTE: 2. If there is NO evidence of electrical arching, inspect the main oil filter (Ref. Step Y.(1)(b)).

3 If cause of pitting can not be determined, metallurgical analysis may be required. Contact the Aircraft OEM.

(b) Inspect the main oil filter:

1 Do a main oil filter patch check (Ref. 79-20-02). The results of the filter patch analysis should be reviewed within the next 25 flight hours. If non allowable debris is found, follow the recommended maintenance actions (Ref. 79-20-02).

2 Regardless of the results of the patch analysis, repeat step Y.(1)(b) every 100 hours, for the next 300 flight hours.

3 If bearing material (AMS 6440/6444 or AMS 6490/6491, (Ref. 79-20-02) is found, remove the engine/gas generator module and send to an approved overhaul facility for repair.

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